# VetNutri MP

Veterinary nutrition manager

## User guide — desktop version

From first steps to the first prescription

Application version: 3.3.18

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Screenshots were taken on a demonstration database, with the French interface. The animals, owners and values shown are fictitious.

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# 1 Before you start

## 1.1 What you will be able to do

By the end of this guide, you will be able to:

- create an animal record and save a consultation;
- choose the nutritional reference suited to the animal;
- build a ration, read its analysis and adjust it;
- prepare a prescription and a report for the owner.

The core of the guide is a worked example followed from start to finish (section 4): Filou, an overweight Labrador. Reproduce it once on your computer: it is the quickest way to get to grips with the application.

## 1.2 The workflow at a glance

Every ration formulation follows the same six steps. Each one builds on the previous one: the ration belongs to a consultation, and the analysis compares the ration with the requirements calculated for that consultation.

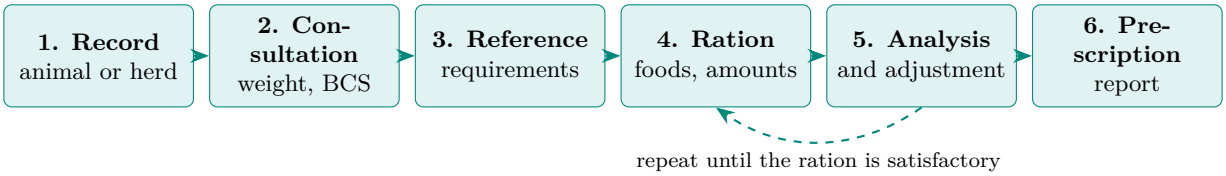



**Figure 1:** The formulation workflow in VetNutri MP.

## 1.3 How to read this guide

- Interface labels are written **like this**, as they appear in the English interface.
- The screenshots show the French interface; the buttons and their positions are the same in English.
- On the screenshots, the markers ❶ ❷ ❸ refer to the explanations that follow the figure.
- Four kinds of boxes help you understand and check:

**Key point**

The essentials to remember.

**Why?**

The nutritional reasoning behind what the application does.

**What you should see**

The expected result on screen. If you see something else, read the step again.

**Caution**

An irreversible action or a common mistake.

VetNutri MP is a decision-support tool. It replaces neither the clinical examination nor the veterinarian's judgement; the veterinarian remains responsible for the data entered and the rations prescribed. Data is stored on your computer and is only sent elsewhere if you export or share it.

# 2 Installation and first launch

On the very first start, the application database is empty. Three one-off actions make it ready to use.

## 2.1 Install the data

1. Start VetNutri MP. A **Data update** window opens.
2. Click **Install update**. Installation takes about a minute.

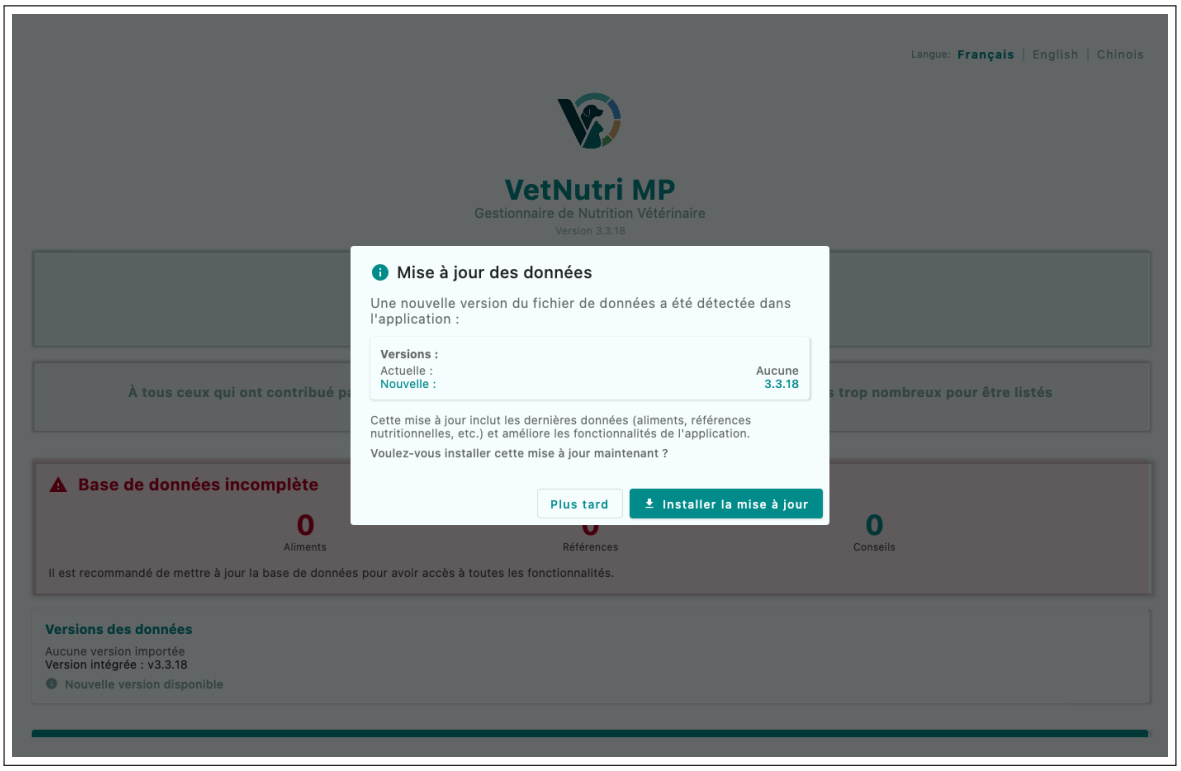



**Figure 2:** Data installation offered at first launch.

**Why?**

The built-in data contains the food database, the requirement references and the advice library. Without it, you could neither choose a reference nor build a ration. The same window reappears when a new version brings more recent data; **Later** postpones the installation.

## 2.2 Accept the terms of use

3. At the bottom of the welcome screen, click **Read and accept terms**.
4. Read the terms, then click **I accept terms**.

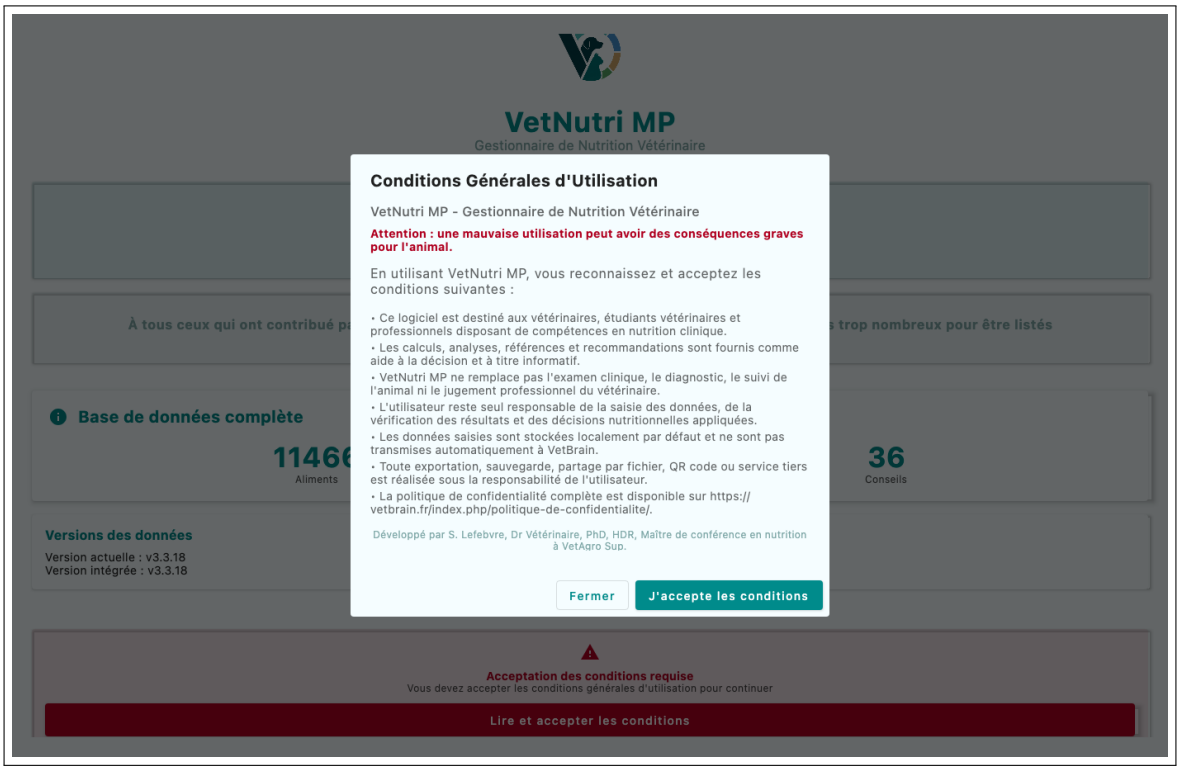



**Figure 3:** Terms of use.

## 2.3 Enter the application

5. Click **Continue**.




**Figure 4:** Welcome screen, ready to use.

1 Database status: number of foods, references and advice items. 2 **Continue** opens the applica-

tion. ③ **Restore a backup** reloads a backup (section 9.6). ④ **Exam mode** is used for supervised exam sessions.

**What you should see**

The box reads **Complete database**, with more than 11,000 foods. On later launches, simply click **Continue**.

**Key point**

The interface language is chosen on this welcome screen (**Language**).

## 3 The main screen

The main screen is the list of animals. Everything starts from here.

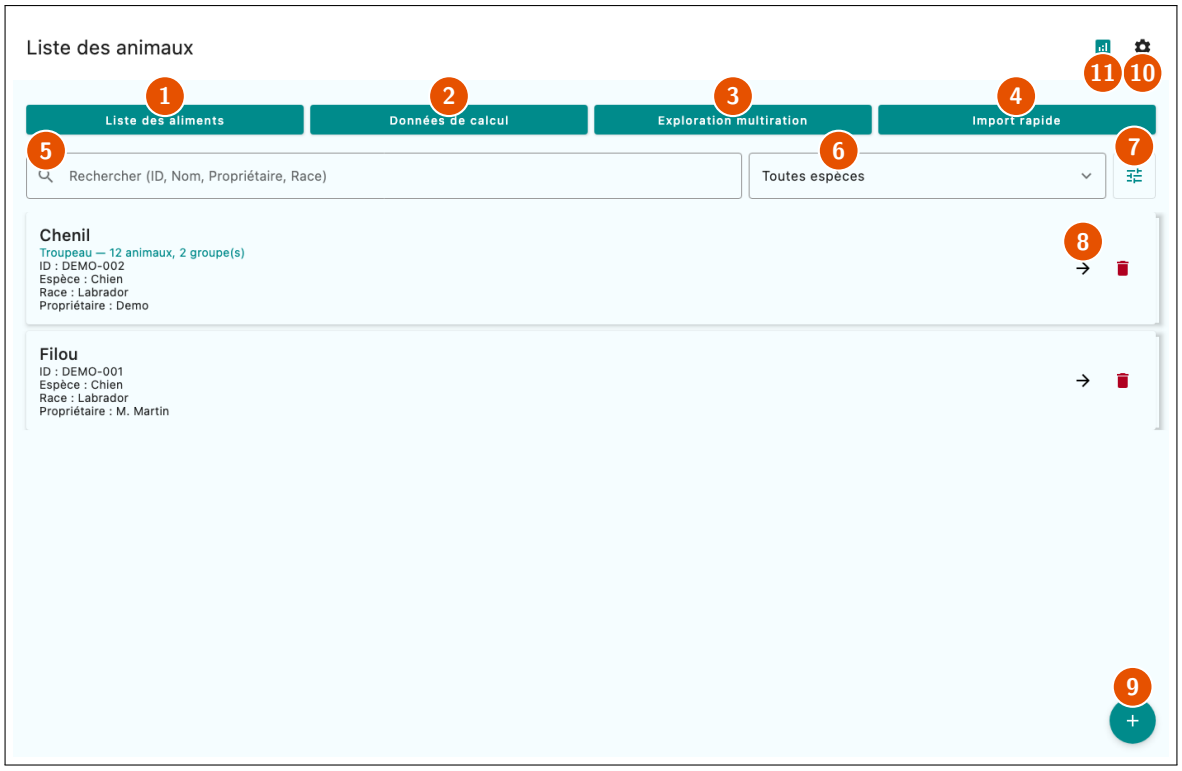



**Figure 5:** The animal list and its controls.

1. **Food list**: browse and edit the food database (section 8.1).
2. **Calculation data**: equations, bibliography, references, advice (section 8.2).
3. **Multi-ration exploration**: calculate rations in series (section 7.2). This button only appears when the feature is enabled.
4. **Quick import**: import animal records.
5. Search by ID, name, owner or breed.
6. Filter by species.
7. Sorting and advanced filters.
8. The arrow opens the animal's record; the red bin deletes it.
9. + creates a new record.
10. Application settings (section 9).
11. **Cross analysis**: compare the consultations of several animals (section 9.7).

# 4 Worked example: Filou's ration, step by step

**The case**

Filou is a 7-year-old intact male Labrador retriever. He weighs 32 kg and his body condition score (BCS) is 7/9: he is overweight. The goal is gradual weight loss with a commercial low-calorie food.
We will create his record, save the consultation, build the ration, correct it using the analysis, then prepare the prescription.

## 4.1 Step 1 — Create the record

**Goal:** record Filou's identity.

1. In the animal list, click + at the bottom right.
2. Leave **Record: Individual** selected.
3. Enter the **ID** (for example the clinic's file number), the **Name** and the **Owner**.
4. Choose the **Species** (*Dog*), then enter the **Breed**, the **Birth date** and the **Sex**.
5. Confirm with the tick button at the bottom right.

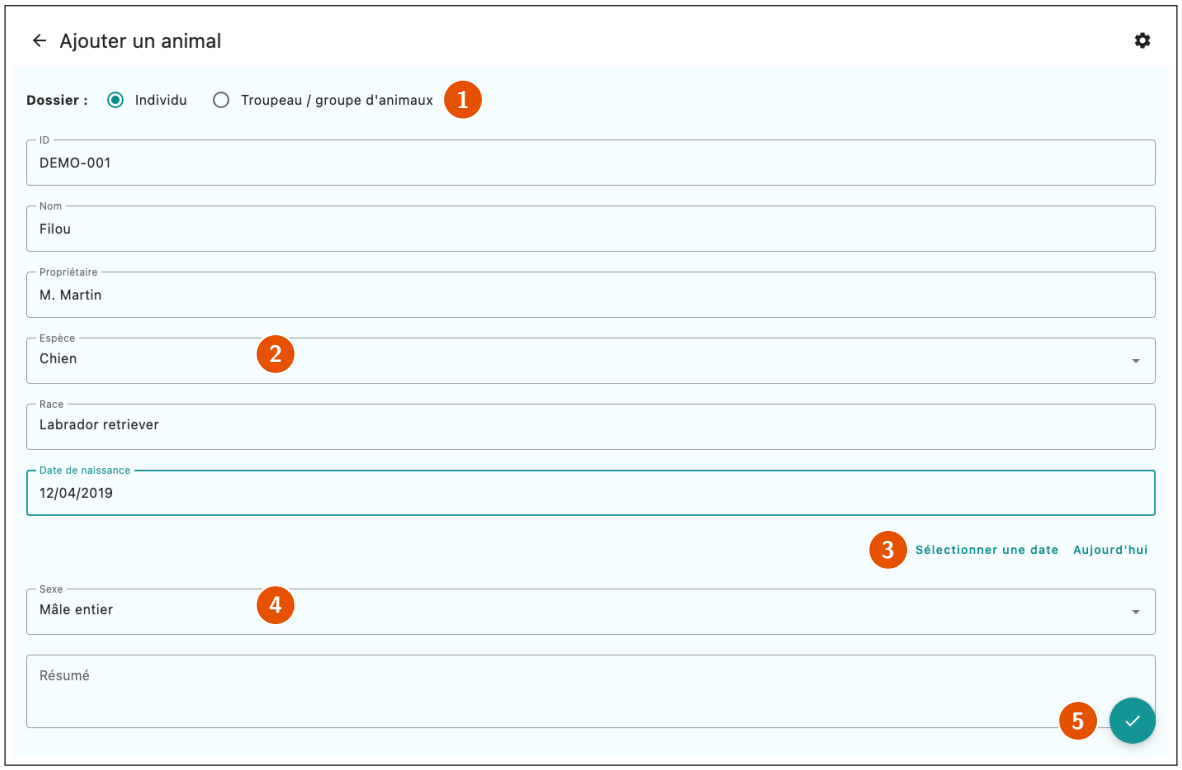



**Figure 6:** The form for creating an individual.

1. **Individual** for a single animal, **Herd / group of animals** for a group.
2. The species: it filters the references and growth curves offered.
3. The birth date: type it, or use **Select a date** or **Today**.
4. Sex and reproductive status.
5. The tick button saves the record.

**Why?**

The species matters: it filters the references and growth curves on offer. Sex and birth date are used for the growth curves and to calculate age. For a group of animals, choose **Herd / group of animals** instead (section 7.1).

## 4.2 Step 2 — Open the animal's record

**Goal:** find the working tabs.

In the list, click the arrow on Filou's row. The record opens on the **Identification** tab.

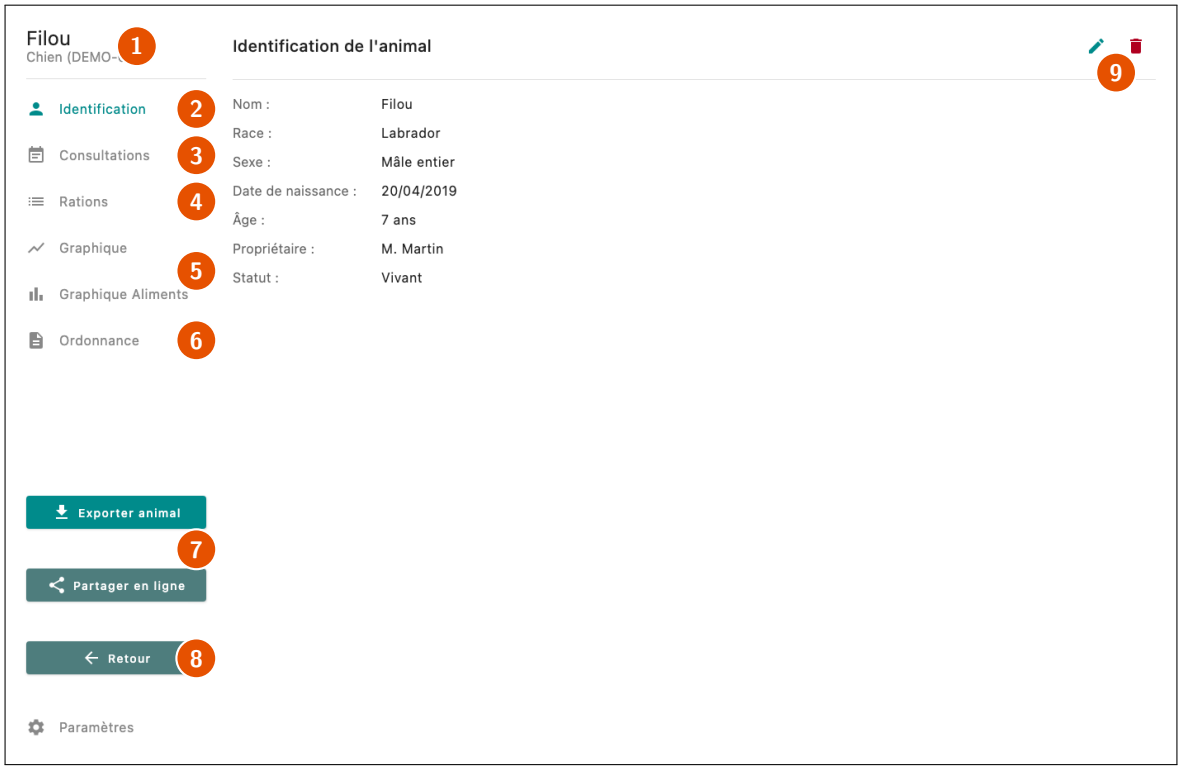



**Figure 7:** The animal's record.

1. Name, species and ID of the open animal.
2. **Identification**: the information entered in step 1.
3. **Consultations**: consultation history (step 3).
4. **Rations**: building and analysing rations (steps 5 to 7).
5. **Graph** and **Food Graph**: weight monitoring and comparisons (section 6.2).
6. **Prescription**: documents for the owner (step 8).
7. **Export animal** and **Share online**: send the record.
8. **Back** to the animal list.
9. Edit or delete the record.

## 4.3 Step 3 — Enter the consultation

**Goal:** describe Filou's condition today.

1. Open the **Consultations** tab, then click + at the bottom right.
2. Check the **Date** 1 and enter the **Weight** 2: 32.

3. Click the three dots of the **Body condition score** 3 and choose *7/9 — Overweight* (figure 9).
4. Read the ideal weight **Estimation** 5: 26.67 kg. If you accept it, enter it in **Ideal weight (kg)** 4.
5. Enter the **Objective** 6: *gradual weight loss*. If needed, add **Keywords** 7 (conditions, context) and an observation.

Do not confirm yet: the reference still has to be chosen (step 4).

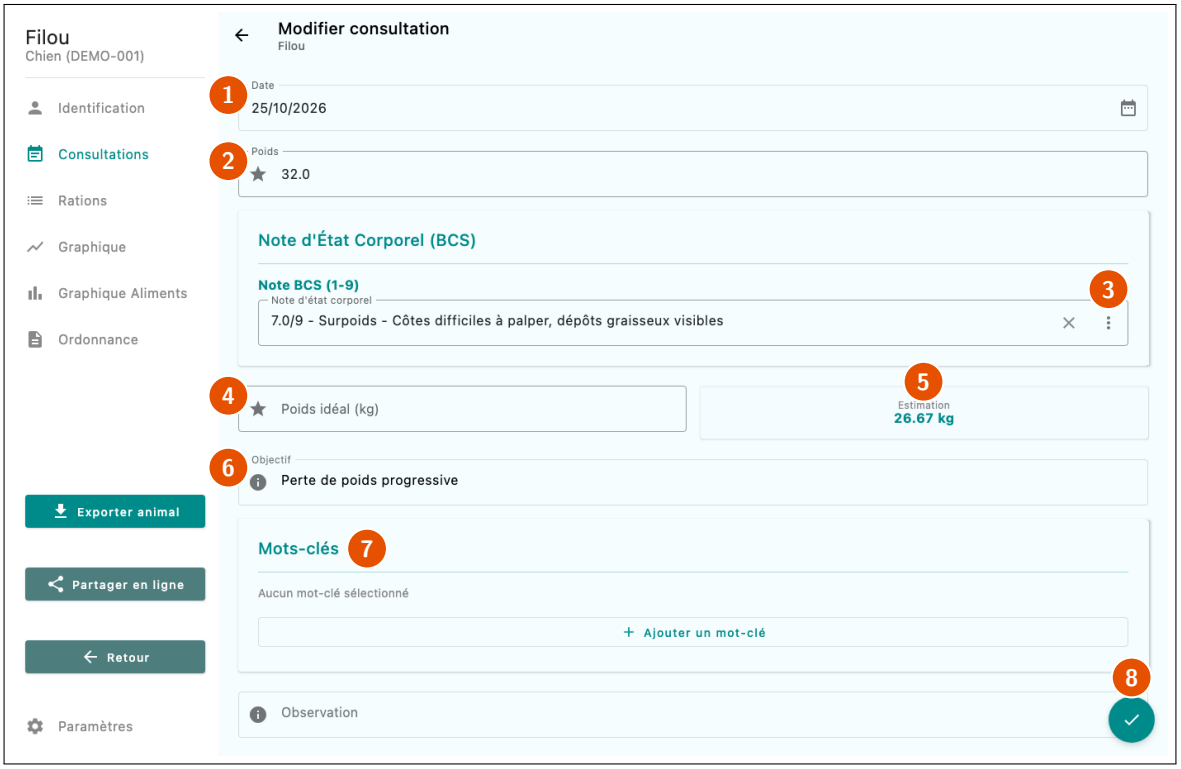



**Figure 8:** Entering the consultation.

8 The tick button saves the consultation (step 4).

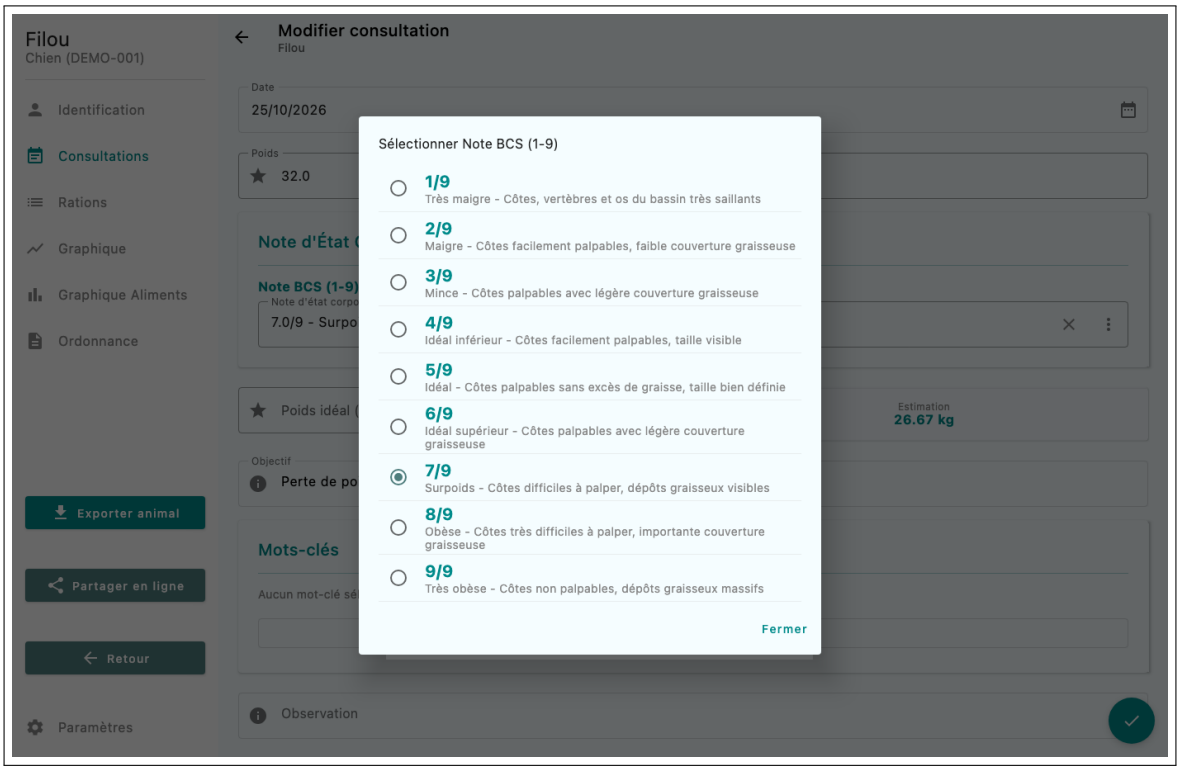



**Figure 9:** Choosing the body condition score out of 9.

**Why?**

The ideal weight is the weight on which requirements are calculated. For an overweight animal, calculating requirements on the current weight would maintain the excess weight.

- The **Estimation** is a suggestion calculated from weight and BCS (section 5).
- It is only used if you copy it into **Ideal weight (kg)**. Otherwise, calculations use the current weight.

## 4.4 Step 4 — Choose the reference

**Goal:** tell the application which requirements to compare the ration with.

1. Scroll the form down to **Nutritional references**.
2. In **General reference**, click the three dots: the list of references for the species opens.
3. Type a few letters in the search box 1 if the list is long, then click the row you want 3: here *Adulte >25kg* (adult over 25 kg).
4. If the animal has a disease to take into account, add it under **Complementary references**.
5. Confirm the consultation with the tick button.

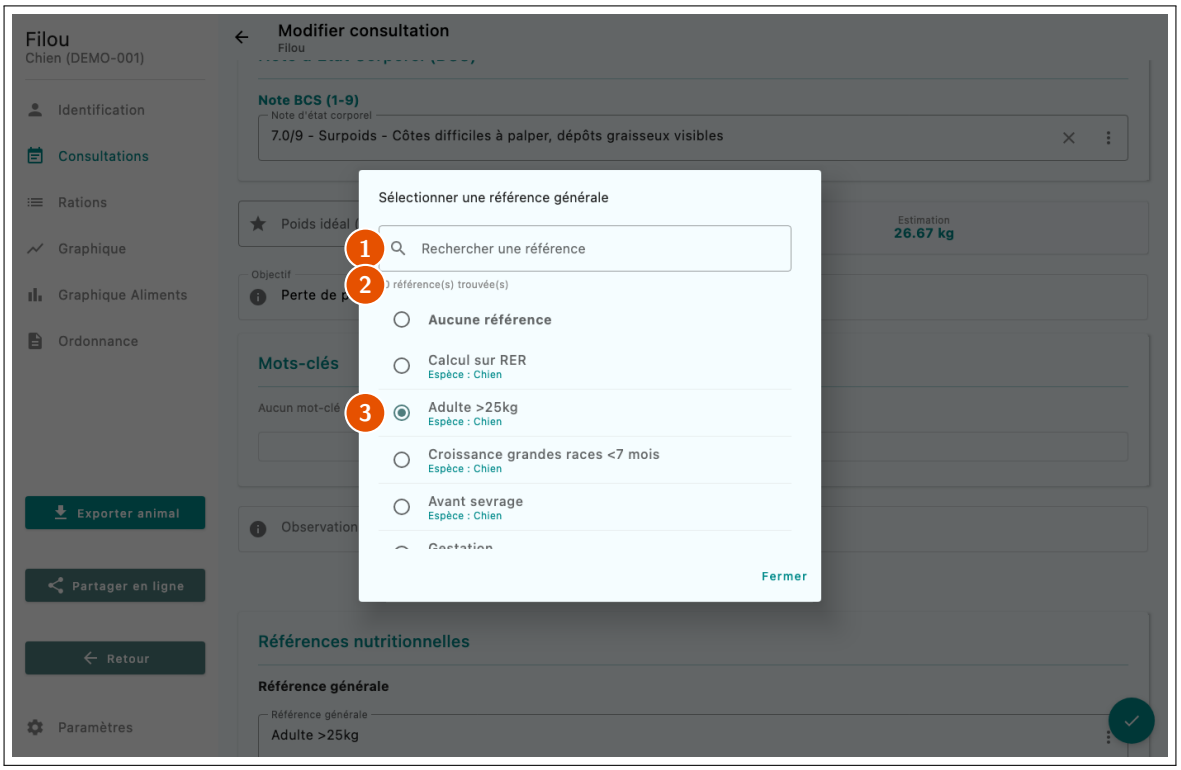



**Figure 10:** Choosing the general reference.

2 The number of references matching the search.

**Why?**

The reference holds the requirements (minimum, optimum, maximum) of each nutrient for one type of animal, as well as the energy requirement equation. It is the yardstick the ration is compared with: a wrong reference gives a wrong analysis, even with a good ration. The **Energy adjustment coefficients** K1 to K5, further down the form, adjust the energy requirement (activity, neutering, environment…).

**Caution**

A consultation without a general reference cannot be saved: the application shows **Missing information**.

**What you should see**

The consultation appears in the list with its date, the weight and **Rations: 1**. The application has created a first, empty ration: the "current ration".

## 4.5 Step 5 — Build the ration

**Goal:** put a food in the ration and set its amount.

1. Open the **Rations** tab. In the **Ration foods** panel, click **+**.
2. Type a name, a brand or an ingredient 1: here *satiety*. Narrow down if needed with the filters 2 (type, indications, species, database).
3. Click a food in the list 3: its details appear on the right 4 (brand, species, indications, ingredients, composition). Check that it is a dog food.
4. Click the round **+** button 5: the food is added with an amount of 100 g. The **+R** button 6 adds it to a new ration instead.

5. Go back to the ration with the arrow at the top left.

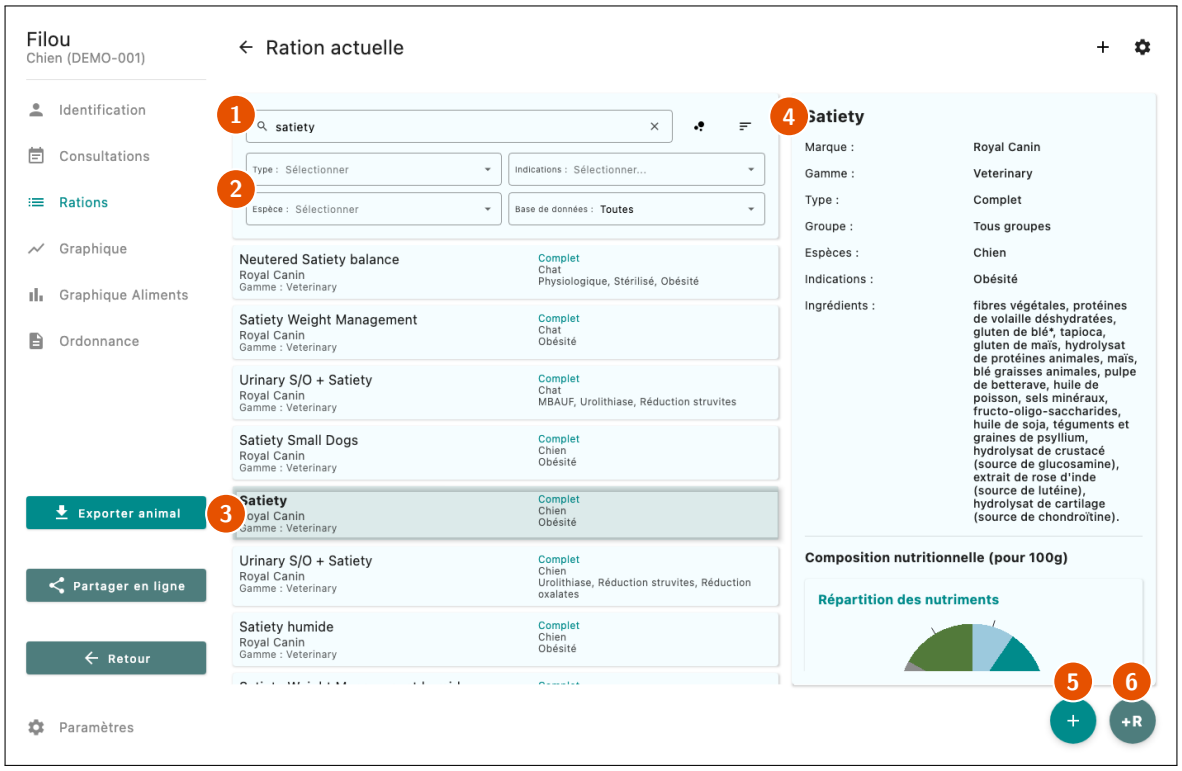



**Figure 11:** Searching for and adding a food.

To set the amount, click the pencil on the food's row, enter the amount in grams (here 380) and confirm with **OK**.

## 4.6 Step 6 — Read the analysis

**Goal:** find out whether the ration meets Filou's requirements.

With 380 g of the food, the **Rations** tab looks like this.

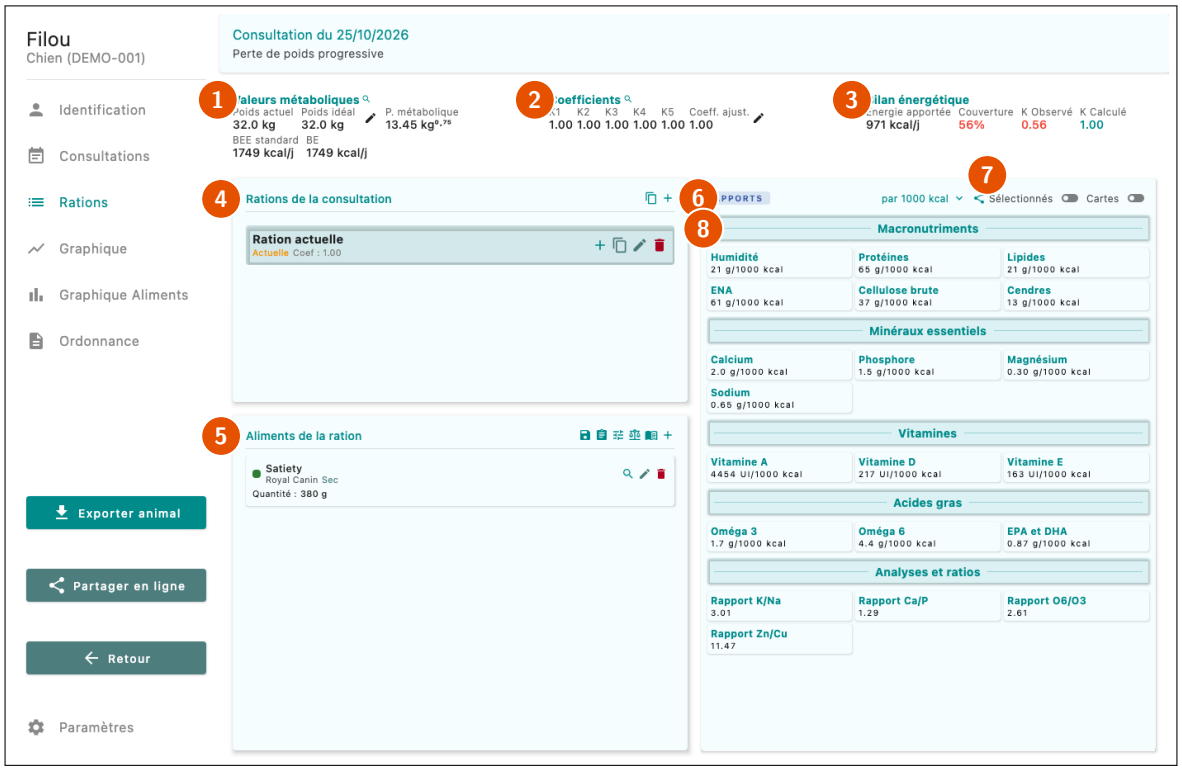



**Figure 12:** Analysis of the 380 g ration.

1. **Metabolic values**: weight, ideal weight, metabolic weight, standard energy requirement (BEE) and energy requirement (BE).
2. **Coefficients** K1 to K5 and the adjustment coefficient. The pencil edits them.
3. **Energy balance**: energy supplied, requirement coverage, observed K and calculated K. **This is the first figure to look at.**
4. The consultation's rations.
5. The foods of the selected ration and the adjustment tools.
6. **INTAKES**: each nutrient, expressed in the reference's unit (here "per 1000 kcal").
7. Expression unit, nutrients shown, PDF export, card or chart view.
8. Nutrients by family. A warning icon flags a deviation from the reference.

### Interpreting the result

Energy coverage is only 56 %: 380 g supply 971 kcal/day for a requirement of 1749 kcal/day. Almost every nutrient is flagged as deficient.

**Why?**

Intakes are related to the animal's energy requirement, not to the energy of the ration (section 5). As long as the ration does not meet the energy requirement, every intake looks low. **So always correct energy first**, then look at the other nutrients.

## 4.7 Step 7 — Adjust the ration

**Goal:** reach an energy coverage close to 100 %.

### Quick adjustment

In the **Ration foods** panel, click the **Quick multi-nutrient adjustment** icon (the scales, to the left of the book and the +). The amounts are recalculated to meet the energy requirement.

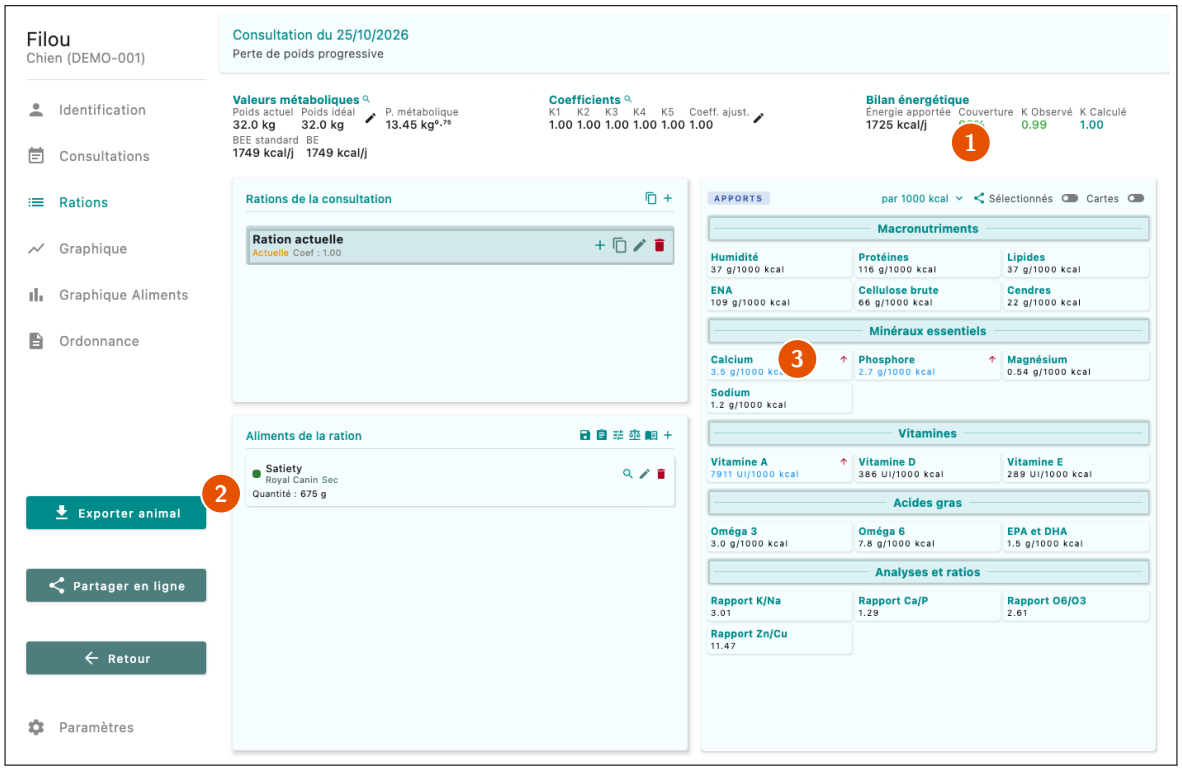



**Figure 13:** The same ration after quick adjustment.

**What you should see**

1 Coverage rises to 99 % and the observed K to 0.99. 2 The amount rises to 675 g. 3 The deficiency warnings disappear; small red arrows now flag a few intakes above the optimum maximum (calcium, phosphorus, vitamin A).

**Why?**

Exceeding the optimum is not necessarily a problem: hover over the icon to read the threshold involved and judge it clinically. For weight loss, energy is set through the ideal weight and the K coefficients, not by feeding less than the calculated ration: otherwise the other nutrients become insufficient.

### Multi-nutrient adjustment

The neighbouring icon opens the full tool, useful when the ration combines several foods.

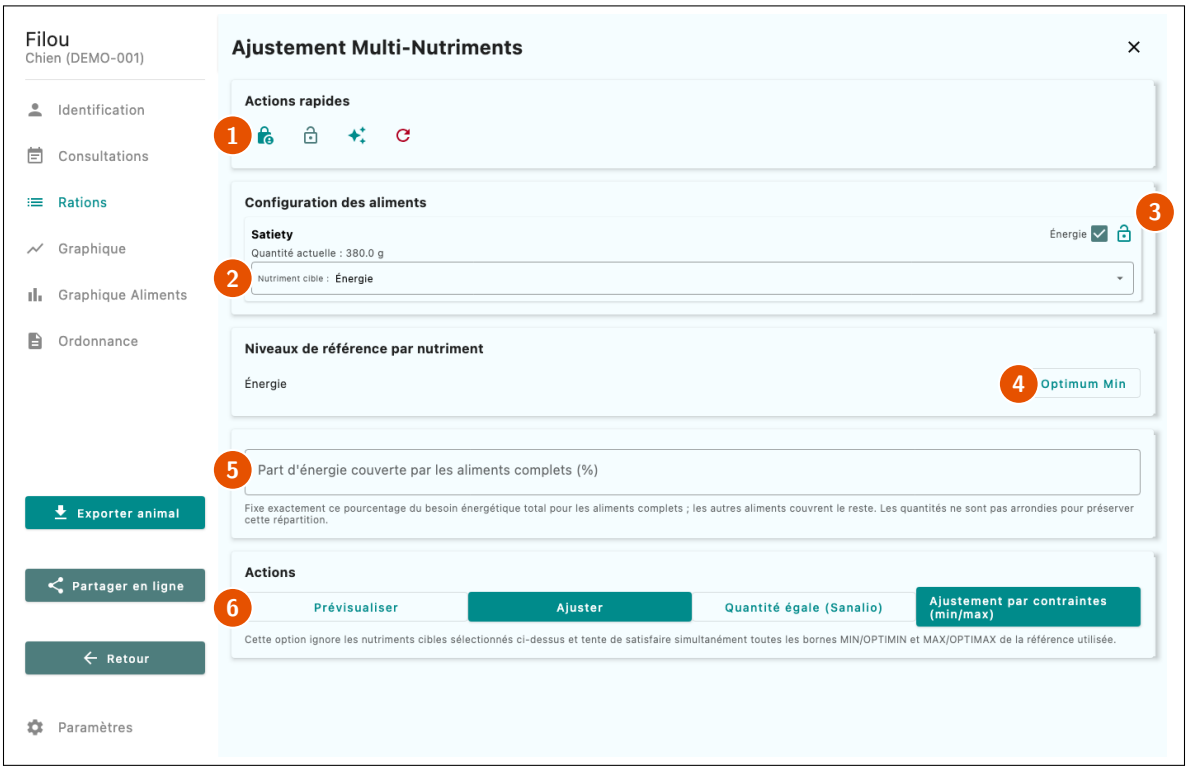



**Figure 14:** Multi-nutrient adjustment.

1. Quick actions: lock or unlock all amounts, automatic adjustment, reset.
2. For each food, the **Target nutrient** it must cover.
3. The padlock freezes a food's amount.
4. The target level for each target nutrient (minimum, optimum…).
5. The share of the energy requirement covered by complete foods, when the ration combines a complete food and supplements.
6. **Preview** without changing anything, **Adjust**, **Equal quantity (Sanalio)**, or **Constraint-based adjustment (min/max)**, which tries to respect all the reference's bounds at once.

### Viewing the ration as a chart

The **Cards** / **Bullet** switch of the **INTAKES** panel shows the composition of the ration, where the energy comes from and, for each nutrient, a bar compared with the reference's thresholds. **Click for more details** opens the detail of a nutrient.

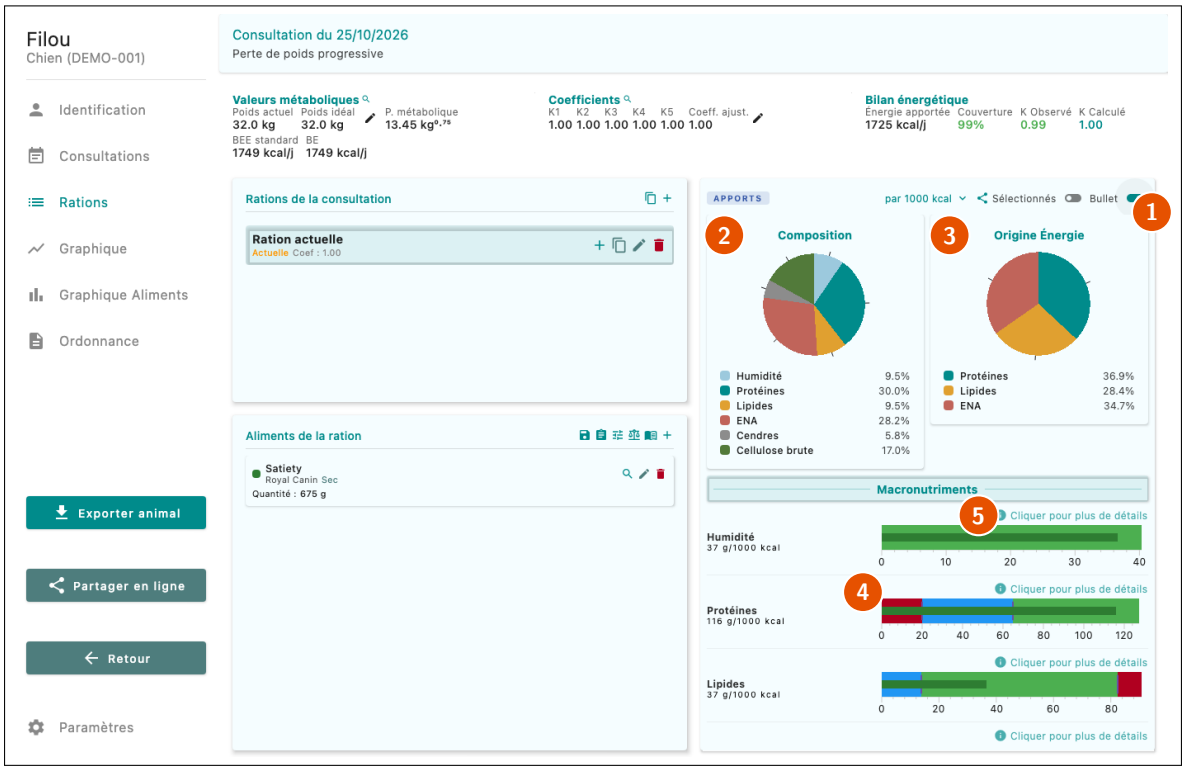



**Figure 15:** Ration analysis in chart view.

1. The **Cards** / **Bullet** switch toggles between the card view and this chart view.
2. **Composition**: the share of each constituent in the ration, by mass.
3. **Energy Origin**: the share of energy supplied by protein, fat and NFE.
4. For each nutrient, a bar places the intake against the reference's thresholds.
5. **Click for more details** opens the nutrient's detail.

## 4.8 Step 8 — Prepare the prescription

**Goal:** give the owner a clear ration and advice.

1. Open the **Prescription** tab.
2. Tick the ration or rations to prescribe 1.
3. Add advice from the library or write a **New section** 2.
4. Fill in the report 3: history, clinical examination, key nutritional factor, additional text.
5. Use the buttons at the bottom 4: **Copy report** (to paste into your practice software), **Preview report**, **Preview prescription**.

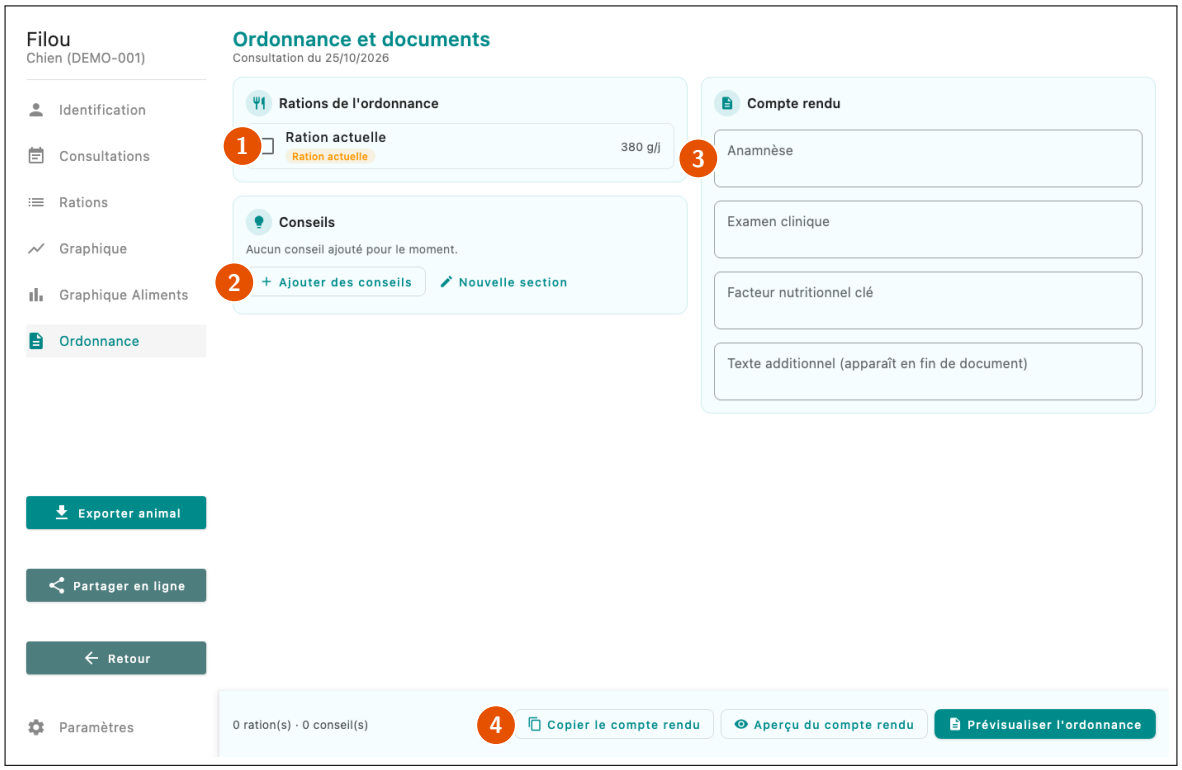



**Figure 16:** Preparing the prescription and the report.

**Key point**

The practitioner's details printed on the prescription are entered once and for all in **Settings** > **Preferences** (section 9).

### 4.9 Summary of the case

| Step | Where | Result for Filou |
|---|---|---|
| 1. Record | List, + | male Labrador, born in 2019 |
| 3. Consultation | **Consultations** | 32 kg, BCS 7/9, estimated ideal weight 26.67 kg |
| 4. Reference | consultation form | *Adulte >25kg* |
| 5. Ration | **Rations**, + | a low-calorie food, 380 g |
| 6. Analysis | **Rations** | 56 % coverage, apparent deficiencies |
| 7. Adjustment | quick adjustment | 675 g, 99 % coverage |
| 8. Prescription | **Prescription** | ration ticked, report, printing |

**Key point**

At a follow-up visit, do not edit the previous consultation: create a new one, or duplicate the previous one with the copy icon. The weight and ration history then stays readable in the graphs.

## 5 Understanding the calculations

This section explains the figures shown in the **Rations** tab, using Filou's values.

## 5.1 The calculation weight

Requirements are calculated on the **ideal weight** if it has been entered, otherwise on the current weight. The estimate offered in the consultation assumes that each BCS point above 5 corresponds to 10 % excess weight:

$$\text{estimated ideal weight} = \frac{\text{weight} \times 100}{100 + (\text{BCS} - 5) \times 10} \qquad \text{Filou: } \frac{32 \times 100}{120} = 26.67 \text{ kg}$$

## 5.2 From weight to energy requirement

1. The **metabolic weight** reflects the fact that requirements do not increase in proportion to weight:
$$W_{\text{met}} = W^{0.75} \qquad \text{Filou: } 32^{0.75} = 13.45 \text{ kg}^{0.75}$$
2. The **standard energy requirement** (BEE) applies the reference's equation. For the reference in the example:
$$\text{BEE} = 130 \times W^{0.75} = 130 \times 13.45 = 1749 \text{ kcal/day}$$
3. The **calculated K** is the product of coefficients K1 to K5 and of the adjustment coefficient. The **energy requirement** (BE) follows from it, increased where relevant by the energy required by disease references:
$$\text{BE} = \text{BEE} \times K \text{ (+ energy from complementary references)}$$

**Key point**

In the example, the ideal weight was not copied over: calculations use 32 kg. With 26.67 kg, the BEE would drop to $130 \times 26.67^{0.75} \approx 1525$ kcal/day.

## 5.3 Energy supplied by foods

A food's energy comes from, in this order:

1. the value entered for the species in the food's record;
2. otherwise the reference's energy density equation: for dogs, the equation for commercial foods (complete or complementary foods), otherwise the Atwater equation (raw foods).

The dog commercial-food equation takes fibre into account:

$$\text{ME (kcal/100 g)} = \left(5.7\,P + 9.4\,L + 4.1\,(\text{NFE} + C)\right) \times \frac{91.2 - 1.43\,\dfrac{100\,C}{100 - H}}{100} - 1.04\,P$$

where $P$ is protein, $L$ fat, $C$ crude fibre and $H$ moisture, in %.

- If crude fibre is not entered, **total dietary fibre** is used instead.
- If nitrogen-free extract (NFE) is not entered, it is calculated: $100 - P - H - C - \text{ash} - L$. The estimated crude fibre is subtracted, so that fibre is not counted twice.

Example: a puppy milk with 33 % protein, 39 % fat, 19.5 % NFE, 3 % moisture and 0.5 % total fibre supplies 541.7 kcal/100 g, i.e. 1083 kcal for 200 g.

### 5.4 The energy balance

$$\text{Coverage} = \frac{\text{energy supplied}}{\text{BE}} \qquad K_{\text{observed}} = \frac{\text{energy supplied}}{\text{BEE}}$$

With 675 g: 1725/1749 = 99 % and $K_{\text{observed}} = 0.99$. The observed K shows which coefficient the ration actually applies; compare it with the calculated K.

### 5.5 Intakes

Intakes are expressed in the reference's unit. In "per 1000 kcal", the daily intake is related to the **standard energy requirement** (BEE), not to the energy of the ration:

$$\text{intake (g/1000 kcal)} = \frac{\text{daily intake (g/day)}}{\text{BEE (kcal/day)}} \times 1000$$

Since the reference's thresholds are expressed the same way, the comparison amounts to comparing the daily intake with the animal's daily requirement. Filou, with 675 g of a food containing 30 % protein: 202.5/1749 × 1000 = 116 g/1000 kcal.

**Key point**

If the ration only covers half of the energy requirement, every intake is half of what it would be with the right amount. Hence the rule: **energy first**.

### 5.6 Warnings

| Message (on hovering over the icon) | Meaning |
|---|---|
| Deficiency: intake below the minimum | below the reference's minimum threshold. |
| Deficiency: intake below the optimal minimum | above the minimum, but below the optimum. |
| Excess: intake above the optimal maximum | above the optimum (red arrow). |
| Incomplete information | the composition of at least one food does not give this nutrient: the actual intake may be higher. |

The **Analyses and Ratios** section gives the K/Na, Ca/P, omega-6/omega-3 and Zn/Cu ratios, calculated on the whole ration and rounded to two decimals (Ca/P = 1.1 / 0.8 = 1.38).

## 6 Going further with rations

### 6.1 Several rations in one consultation

A consultation can hold the owner's current ration and one or more proposals, to compare them.

- **+** in the **Consultation rations** panel adds a ration; the icon next to it reuses a ration from a previous consultation.
- On each ration: add a step to the progressive plan (food transition), duplicate, edit, delete.
- In **Ration foods**: save the ration as a recipe, copy it, open the saved recipes.

## 6.2 Monitoring graphs

The **Graph** tab offers four views: **Weight evolution**, **Energy distribution**, **Energy density of rations** and **Ration nutrients**.

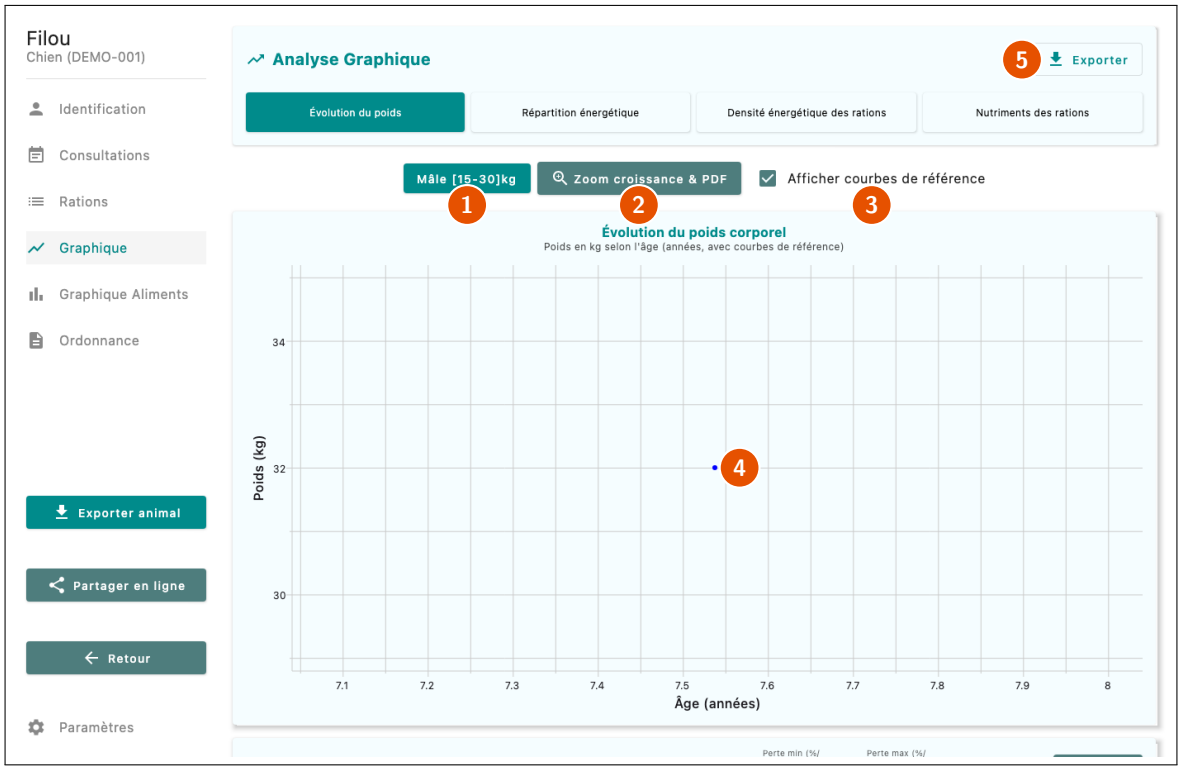



**Figure 17:** Weight evolution and reference curves.

1. The suggested reference curve (sex and weight class); click to choose another one.
2. Enlarging the growth curves and exporting them to PDF.
3. Show or hide the reference curves.
4. Each weighing, plotted against age.
5. **Export**: the ration charts as images, with a Markdown or LaTeX document listing the foods of each ration.

The weight view plots each weighing against age. Below the graph, the **Weight history** lists the animal's weighings; the + button adds one, and the fields next to it set the weight-loss target (minimum and maximum loss in % per week, target weight).

**Why?**

The reference curve offered by default matches the animal's sex and weight class. That class is set, in this order:

1. by the expected adult weight, for a puppy (variable AW of the consultation);
2. otherwise by the ideal weight, entered or estimated from the BCS;
3. otherwise by the latest weight.

An overweight animal is therefore not placed in too high a class. The button above the graph lets you choose another curve.

In the ration views, each ration has a number, repeated in the legend below the graph: 1, 2… for the rations of the first consultation, 101, 102… for the next one. Two options control the display:

- **Current consultation only** limits the graph to the consultation on display;
- **Area of possibilities and sums** outlines the rations of each consultation (orange for current

rations, blue for proposals) and adds their mean weighted by the coefficients, marked Σ. Ration numbers do not change.

The **Food Graph** tab compares foods with each other: add them with + in the list on the left, then click **View graphic analysis**.

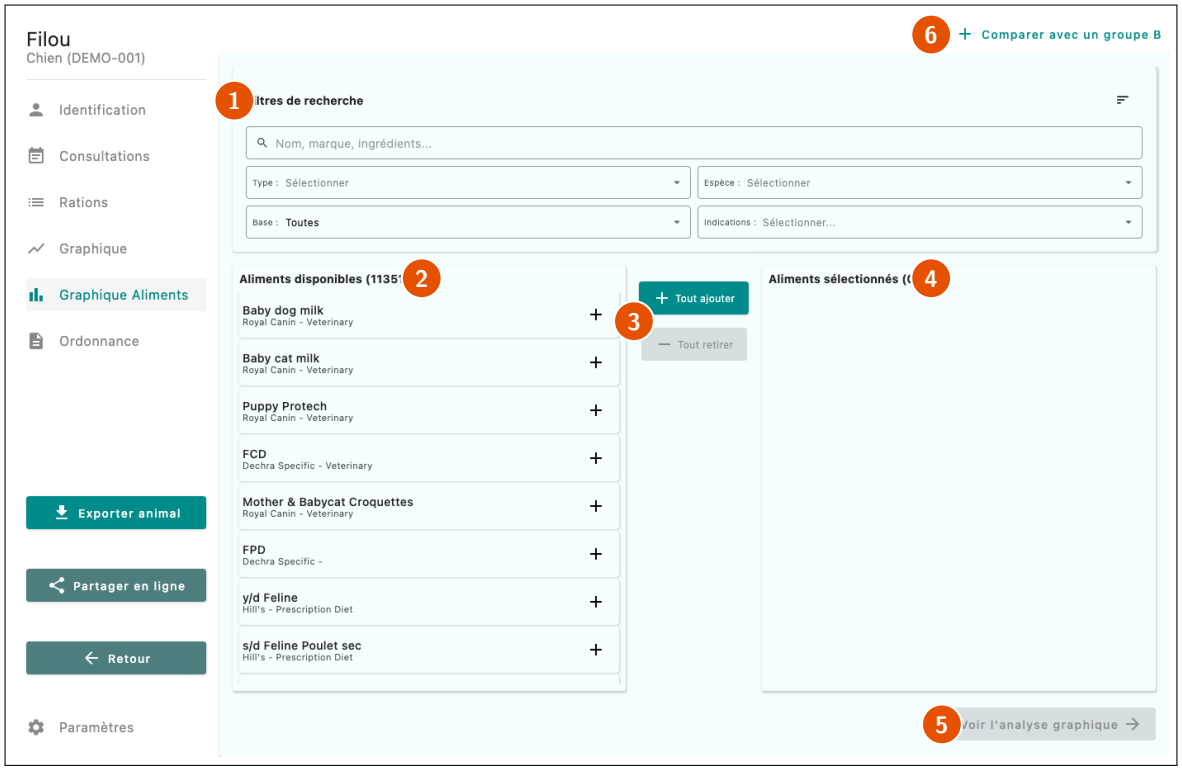



**Figure 18:** Selecting foods to compare.

1. Search and filters (type, species, database, indications).
2. The foods found; the + on a row adds the food to the comparison.
3. **Add all** and **Remove all**.
4. The selected foods.
5. **View graphic analysis** opens the comparison charts.
6. **Compare with a group B** adds a second selection: the two groups are compared with box plots (energy density, protein…), with export.

### 6.3 Sending a record

From the record's left-hand menu:

- **Export animal** saves the complete record (consultations, rations, foods used) to a file that can be imported on another computer;
- **Share online** sends the record through an online service.

## 7 Special cases

### 7.1 A herd

**When to use it?** To work out a ration for a whole facility or group of animals: a kennel, a cattery, a batch of farm animals.

When creating the record, choose **Herd / group of animals**. A herd has no species, breed or sex of its own: it is described by groups of individuals (name, number, mean weight, species, breed, sex). It can mix several species; each group's reference and K are chosen in each consultation.

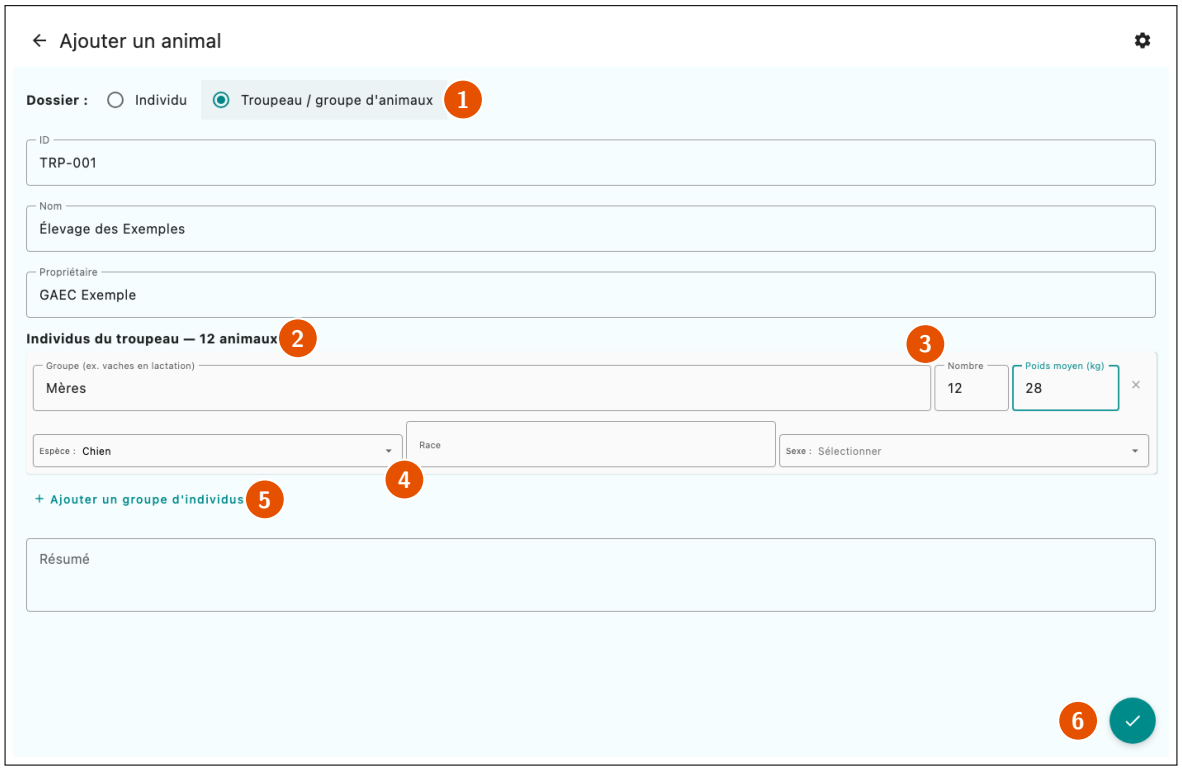



**Figure 19:** Creating a herd of twelve individuals.

1. **Herd / group of animals** is selected.
2. The herd's total number of animals, the sum of the groups.
3. For each group: its name, the number of animals and their mean weight; the cross removes the group.
4. The group's species, breed and sex.
5. **Add a group of animals**; a new group takes the species of the previous one.
6. The tick button saves the herd.

### The consultation

The **Herd status** replaces the individual's weight, BCS, reference and K. For each group, enter the number of animals and their mean weight on the date of the consultation, the mean body condition score, the reference (for the group's species) and the group's K, together with any variables of the reference's equations. The requirement of one animal is the reference's standard requirement multiplied by the group's K.

### The ration and its analysis

A herd's rations are entered as daily amounts for the whole herd. The **Rations** tab shows the herd panel above the rations.

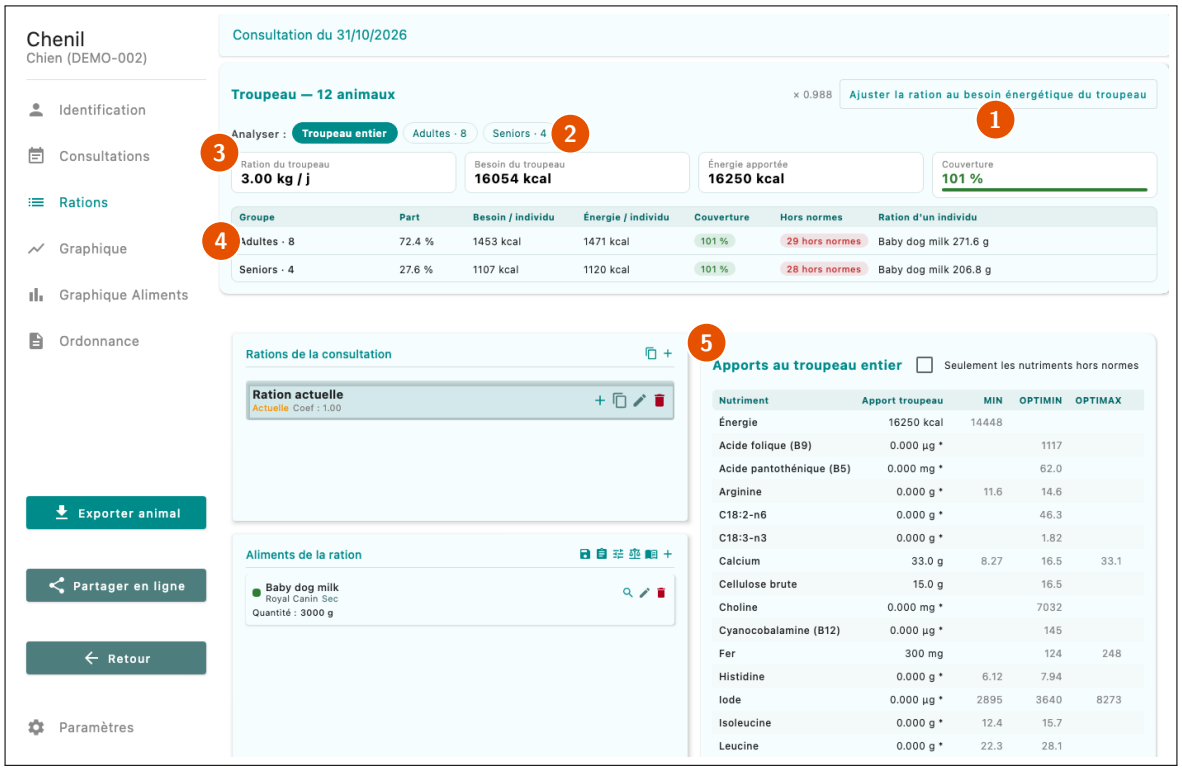



**Figure 20:** The ration of a herd of twelve dogs in two groups.

1. **Adjust the ration to the herd's energy requirement** multiplies every amount by the factor shown (here ×0.988) so that the energy supplied equals the herd's requirement.
2. What is analysed: **Whole herd**, or one group (its name and number of animals).
3. The herd's daily ration, the herd's requirement, the energy supplied and the coverage (green from 90 to 110 %, orange up to 80 and 120 %, red beyond).
4. One row per group: share of the ration, requirement and energy of one animal, coverage, number of out-of-range nutrients and ration of one animal.
5. **Whole-herd intakes**, nutrient by nutrient.

**Why?**

The herd ration is shared between the groups in proportion to their energy requirement: one animal of group $i$ receives

$$\text{amount} \times \frac{B_i}{\sum_j n_j B_j}$$

where $n_j$ is the number of animals in group $j$ and $B_j$ the energy requirement of one of them. In the example, a 25 kg adult receives 271.6 g and a 20 kg senior (K = 0.9) 206.8 g.

**Analysing a group.** Click a group's row (or its chip): the ration shown becomes that of one animal of the group, analysed like a single animal's ration, with the group's weight, reference and K. This ration is read-only; click the row again, or **Whole herd**, to return to the herd ration and edit it.

**The whole herd.** The **Whole-herd intakes** table compares the total daily intake with the sum of the animals' requirements (number × requirement of one animal, group by group), for each nutrient: energy first, then out-of-range nutrients, then the others in alphabetical order. **Only out-of-range nutrients** shortens the list. An asterisk marks an incomplete value: the composition of at least one food does not give this nutrient.

**Caution**

A compliant total can hide a deficiency in one group: an over-fed group makes up, in the total, for a group that falls short. Always check the last column, **Out-of-range groups**, and the number of deviations of each group in the upper table.

## 7.2 Multi-ration exploration

**When to use it?** To prepare standard rations, for example for a whole range of weights, rather than a ration for one specific animal.

First enable it in **Settings** > **Administration** > **Enable multi-ration analysis**; the **Multi-ration exploration** button then appears in the animal list.

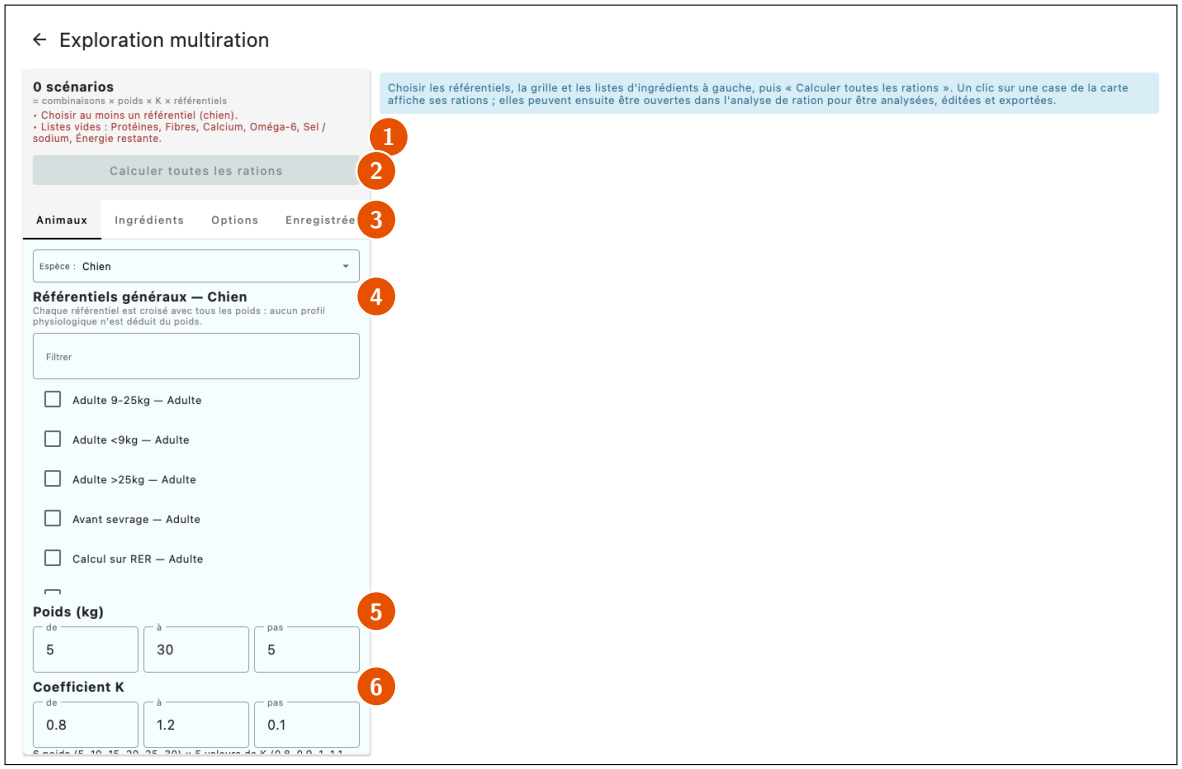



**Figure 21:** Setting up a multi-ration exploration.

1. The number of scenarios to calculate and what still blocks the calculation.
2. **Calculate all rations**, active once the configuration is complete.
3. The tabs: **Animals**, **Ingredients**, **Options**, **Saved**.
4. The species and the general references to combine.
5. The weight grid (from, to, step).
6. The K coefficient grid (from, to, step).

1. Choose the species and tick one or more general references.
2. Define the grid: weight (from, to, step) and K coefficient (from, to, step). The number of scenarios is shown.
3. In the **Ingredients** tab, build the lists of foods supplying protein, fibre, calcium, omega-6, salt and energy. The adjustment follows this order.
4. Click **Calculate all rations**. Clicking a cell of the map shows its rations, which can be opened in the usual analysis to edit and export them.

The **Saved** tab keeps named configurations; results are recalculated each time they are opened.

# 8 Managing data

## 8.1 The food database

**Food list** (main screen) shows the whole database, with the same filters as the food search of a ration. The pencil edits a food, the bin deletes it, + creates one.

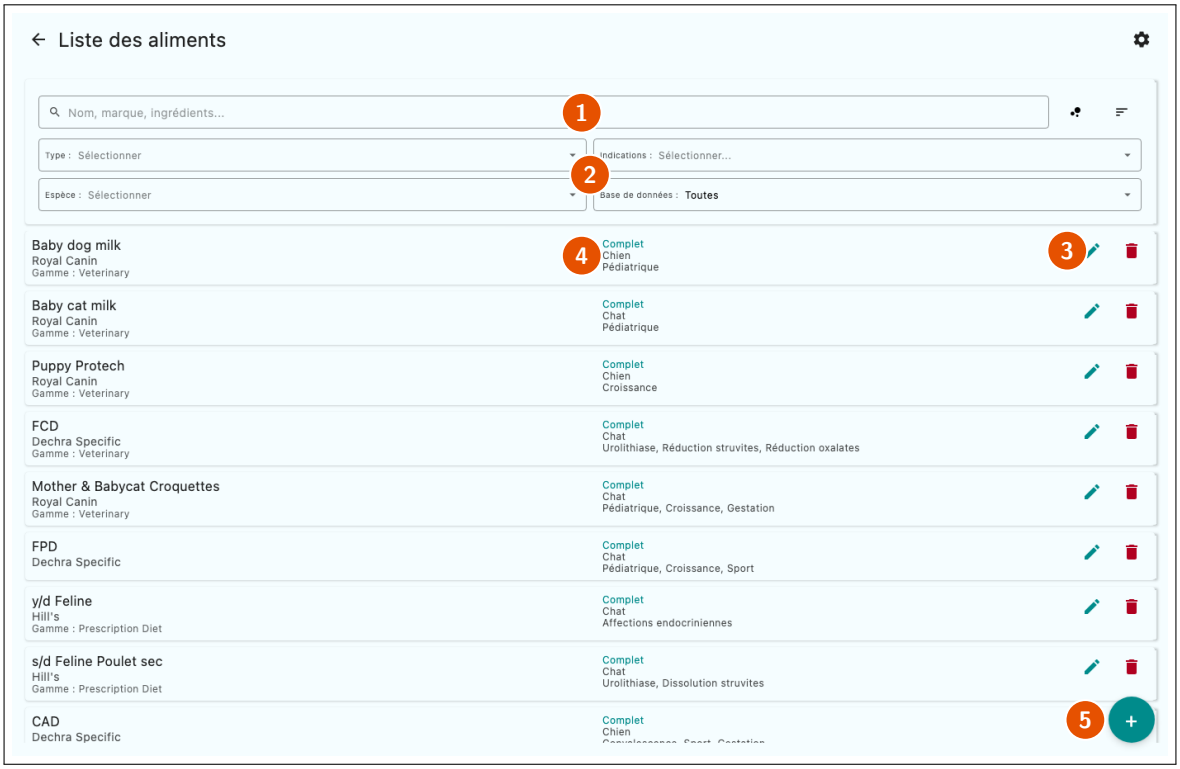



**Figure 22:** Food list.

1. Search by name, brand or ingredient.
2. Filters: type, indications, species, database.
3. On the right of each row, the pencil edits the food, the bin deletes it.
4. Food type, species and indications.
5. + creates a food.

A food record has three tabs: **General information** (name, brand, range, database, price, packaging, type), **Nutritional Composition** and **Bibliography**. Save with the tick button at the bottom right.

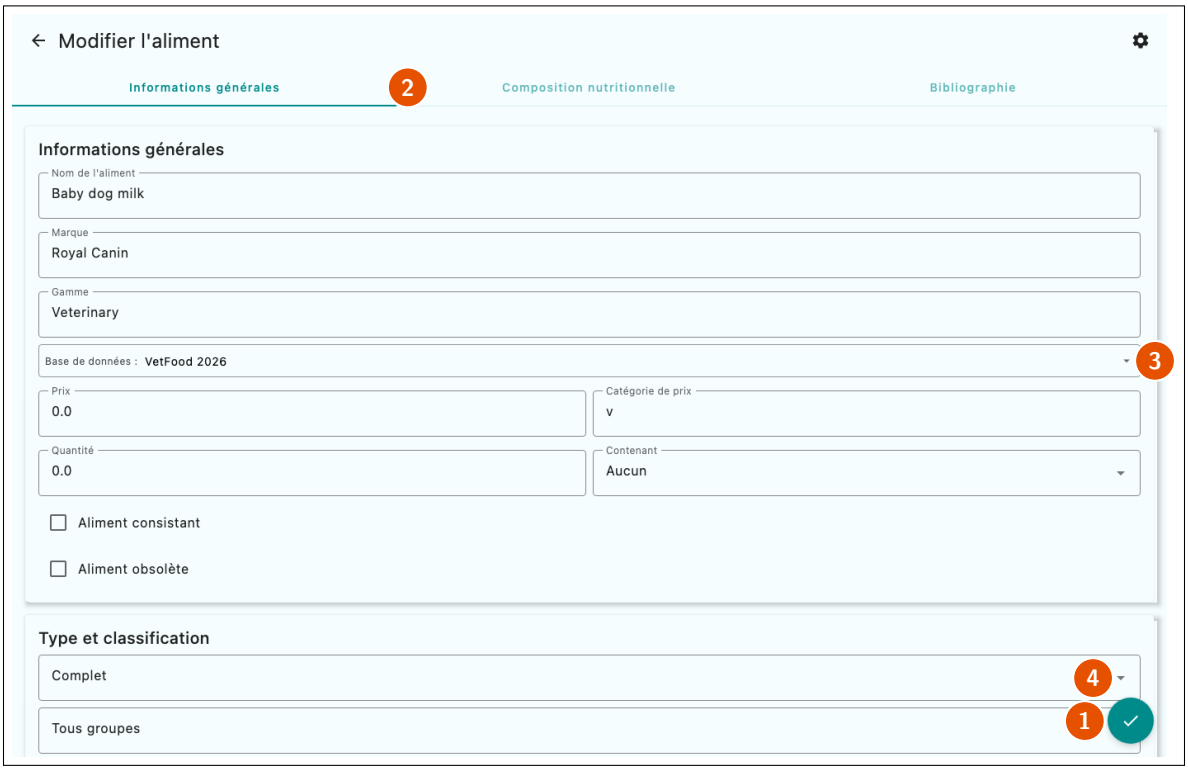



**Figure 23:** Editing a food.

1. The tick button records the changes.
2. The three tabs: **General information**, **Nutritional Composition**, **Bibliography**.
3. The food's source database.
4. The food type: complete or complementary (energy from the commercial-food equation), otherwise raw food (Atwater equation), see section 5.

**Key point**

An incomplete composition produces "Incomplete information" warnings in the analysis. For a food you create, enter at least energy, macronutrients, calcium and phosphorus.

## 8.2 Calculation data

The **Calculation data** button gives access to the elements the calculations rely on. They ship with the application. Edit them with care: any change alters the analyses of every consultation that uses them.

| Tab | Content |
|---|---|
| **Equations** | formulas for energy requirement, metabolic weight and energy density (for example `130*BW^0.75`) |
| **Bibliography** | references supporting the requirement references and food values |
| **Calculation systems** | the requirement references: by species, life stage or disease |
| **Custom advice** | advice library reused in the prescription |

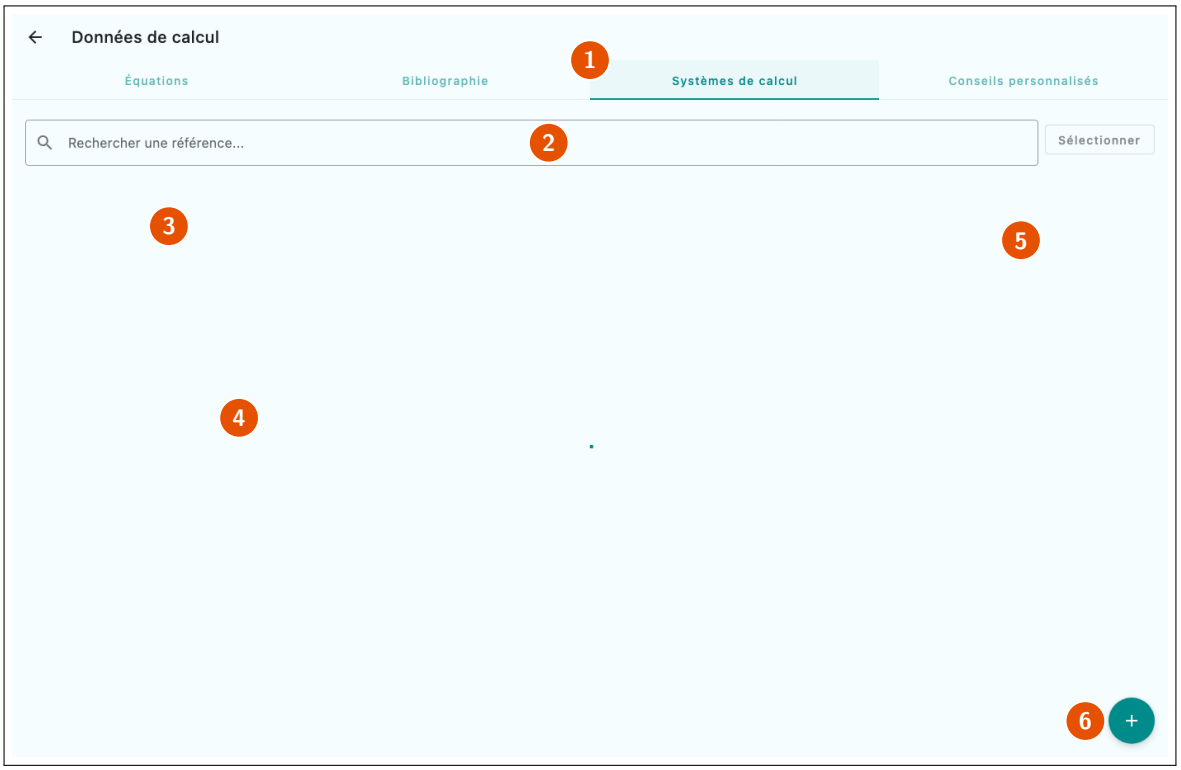



**Figure 24:** Nutritional references (Calculation systems tab).

1. The four tabs of the calculation data.
2. Searching for a reference.
3. Name, species and physiological stage of the reference.
4. A red line marks a disease reference, to be added as a complementary reference.
5. Edit, duplicate or delete the reference.
6. + creates a reference.

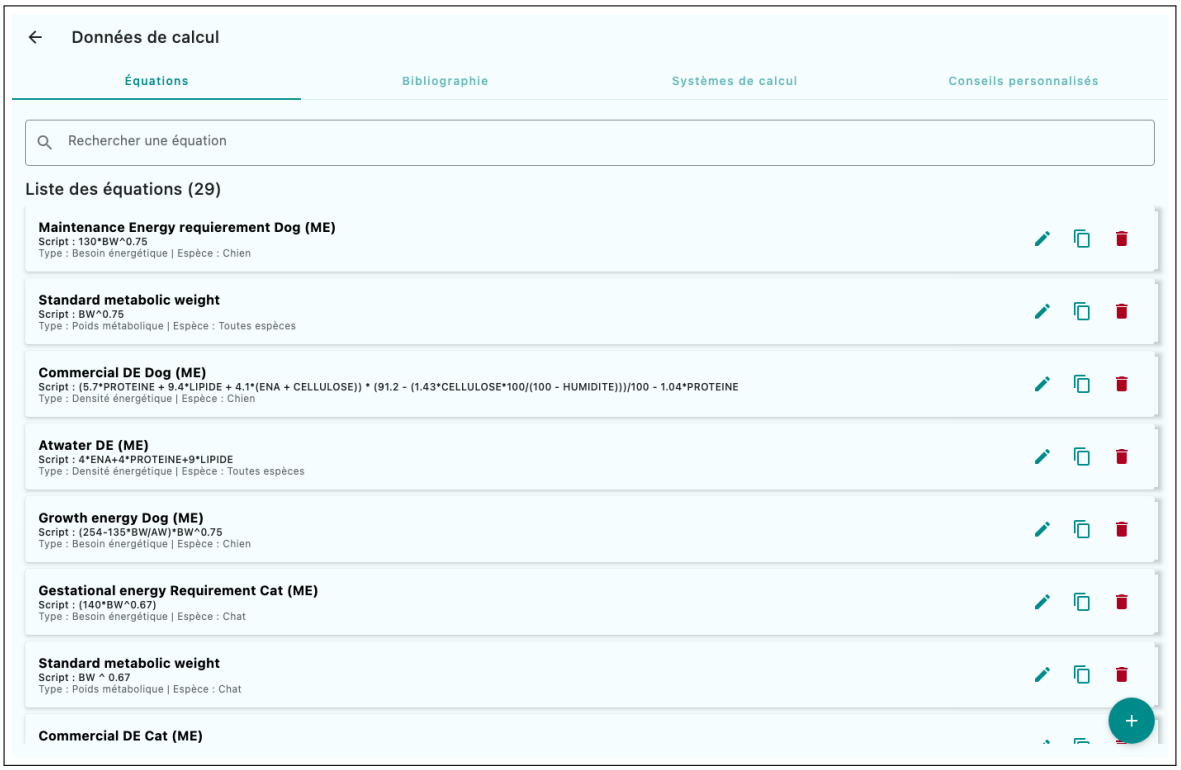



**Figure 25:** Equations.

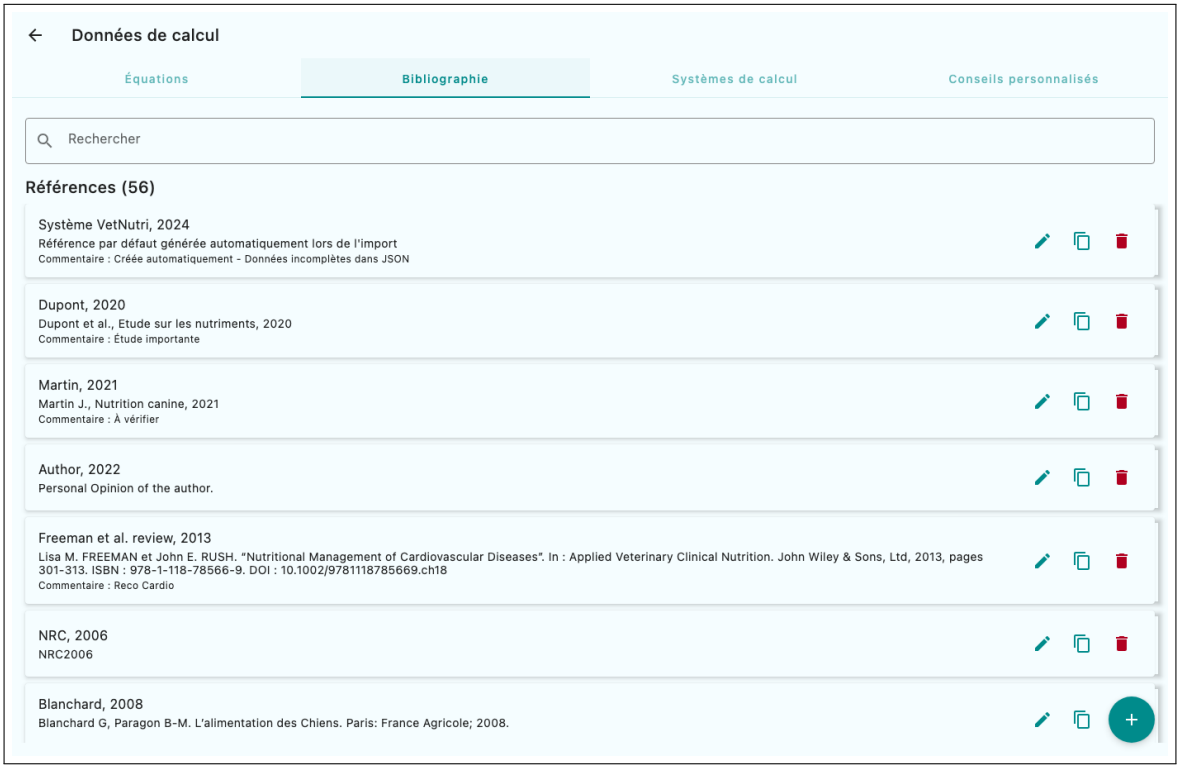



**Figure 26:** Bibliography.

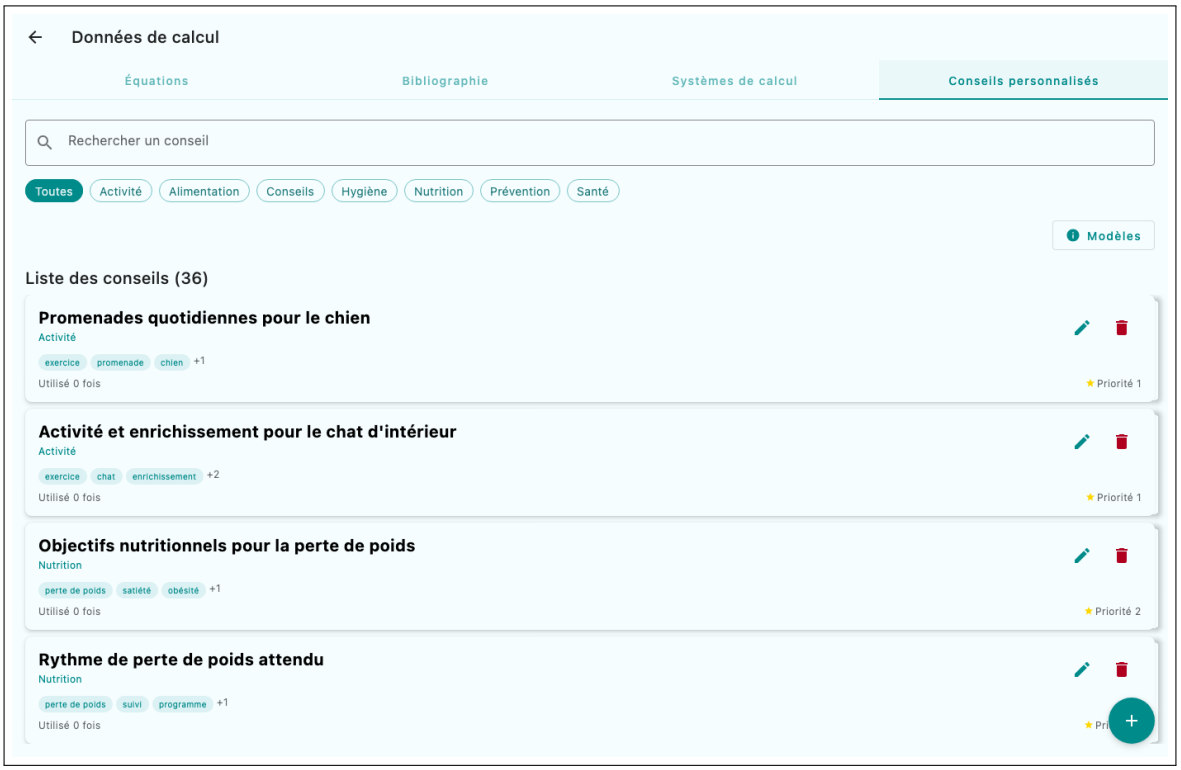



**Figure 27:** The advice library.

# 9 Settings

The gear icon on the main screen opens the settings, in six tabs.

| Tab | Purpose |
|---|---|
| **Interface** | interface size (50 to 200 %) |
| **Preferences** | practitioner's details for the prescription; unit used to express requirements, by species |
| **Import/Export** | data exchange (animals, foods, recipes, equations, references, advice) |
| **Excel** | export and import of foods as Excel-compatible CSV |
| **Recipes** | recipes saved from rations |
| **Administration** | multi-ration, re-import of the initial data, backups, VetNutri 2 import, maintenance |

## 9.1 Preferences: to do right after installation

Enter your details (name, registration number, address, phone, e-mail), printed on the prescription, then click **Save**. The **Requirement expression by species** section sets the unit in which requirements are expressed (for example "per 1000 kcal").

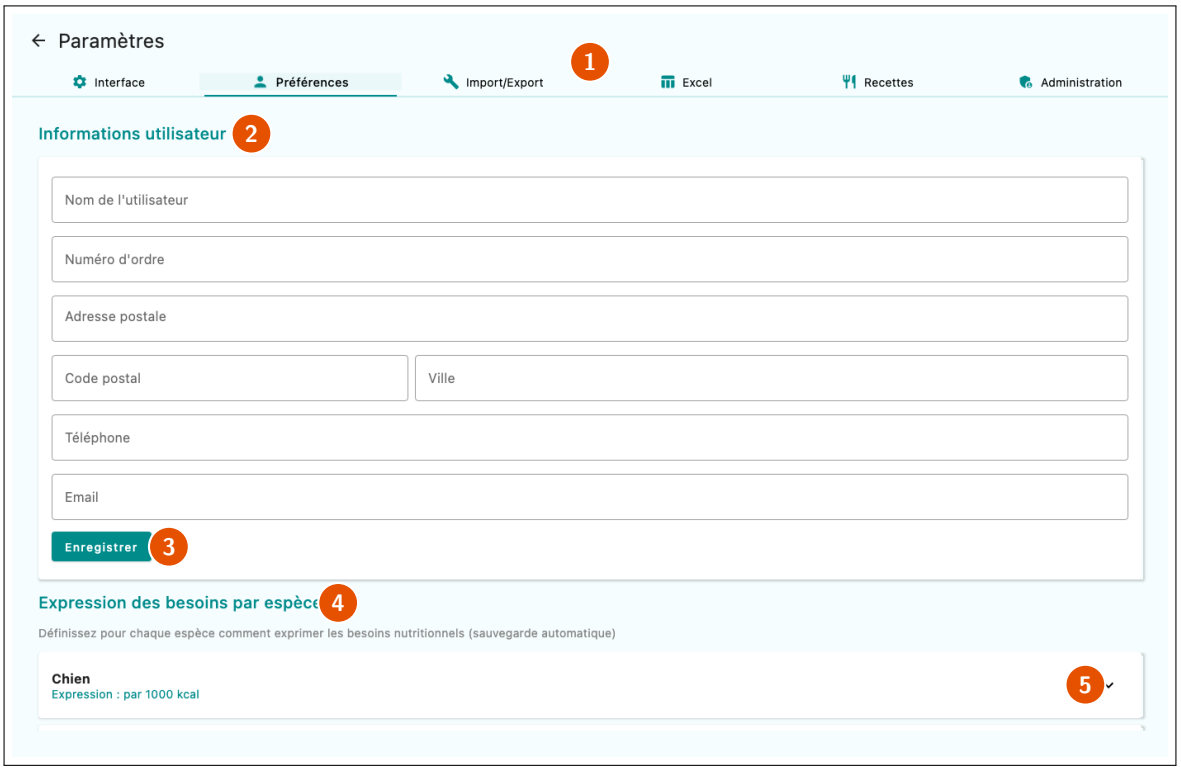



**Figure 28:** Preferences.

1. The six settings tabs.
2. The practitioner's details, printed on the prescription.
3. **Save** the details.
4. **Requirement expression by species**: the unit in which requirements are shown.
5. Click a species to change its unit (for example "per 1000 kcal").

## 9.2 Interface

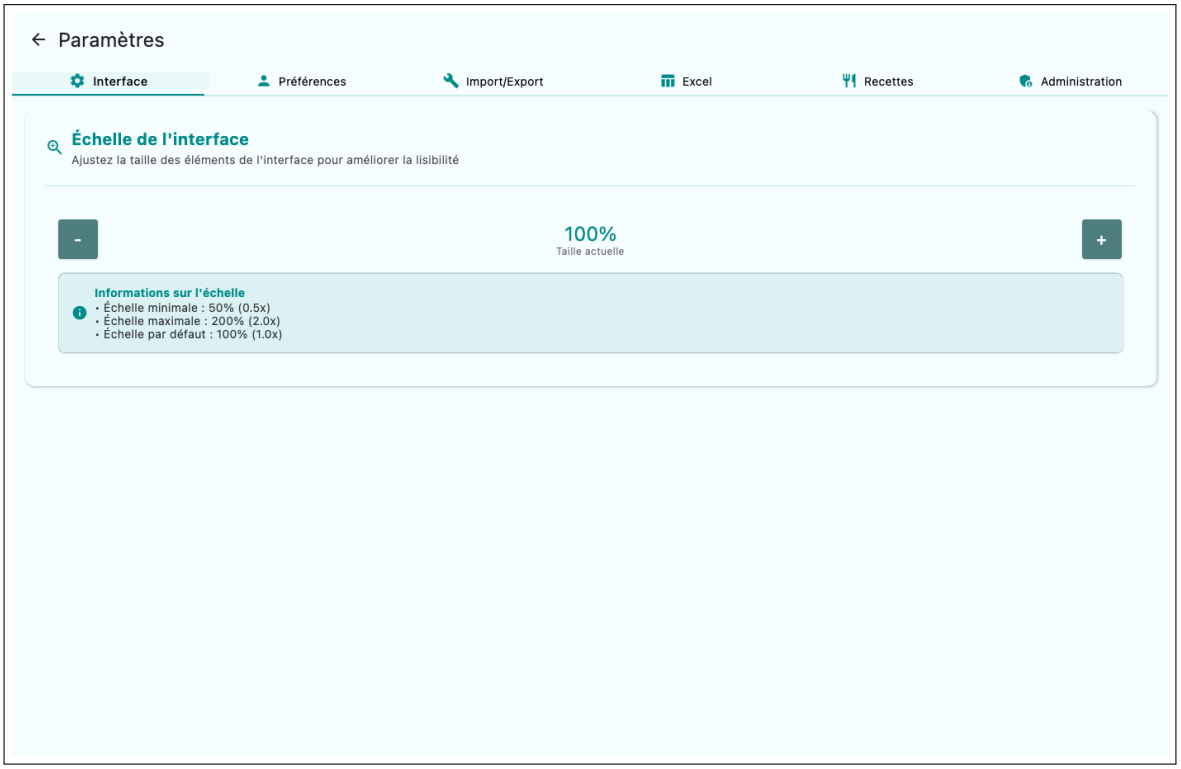



**Figure 29:** Interface size.

## 9.3 Import, export and Excel

The **Import/Export** tab exports and imports data in API format, with advanced selection, and imports foods, animals or nutritional references (`.vbnr.json`).

In a references file, each requirement states its unit: per Mcal (per 1000 kcal), per kg of body weight, per kg of metabolic weight, as an absolute value or as a ratio. A requirement whose unit is not recognised (for example “DM”) is not imported: it is listed in the import report, so that you can correct the file.

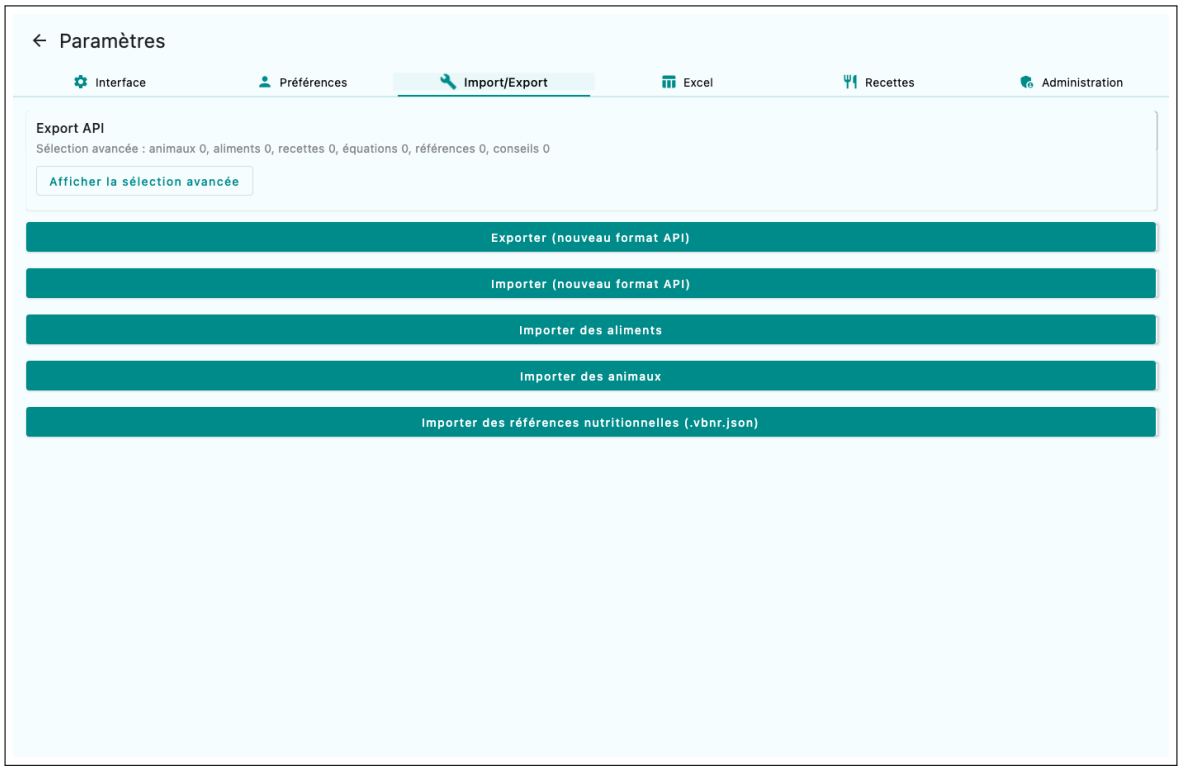



**Figure 30:** Data import and export.

The **Excel** tab exports and imports foods as CSV: semicolon separator, UTF-8 encoding, one header row, one column per nutrient with the unit in the header. A preview lets you check the data before confirming the import.

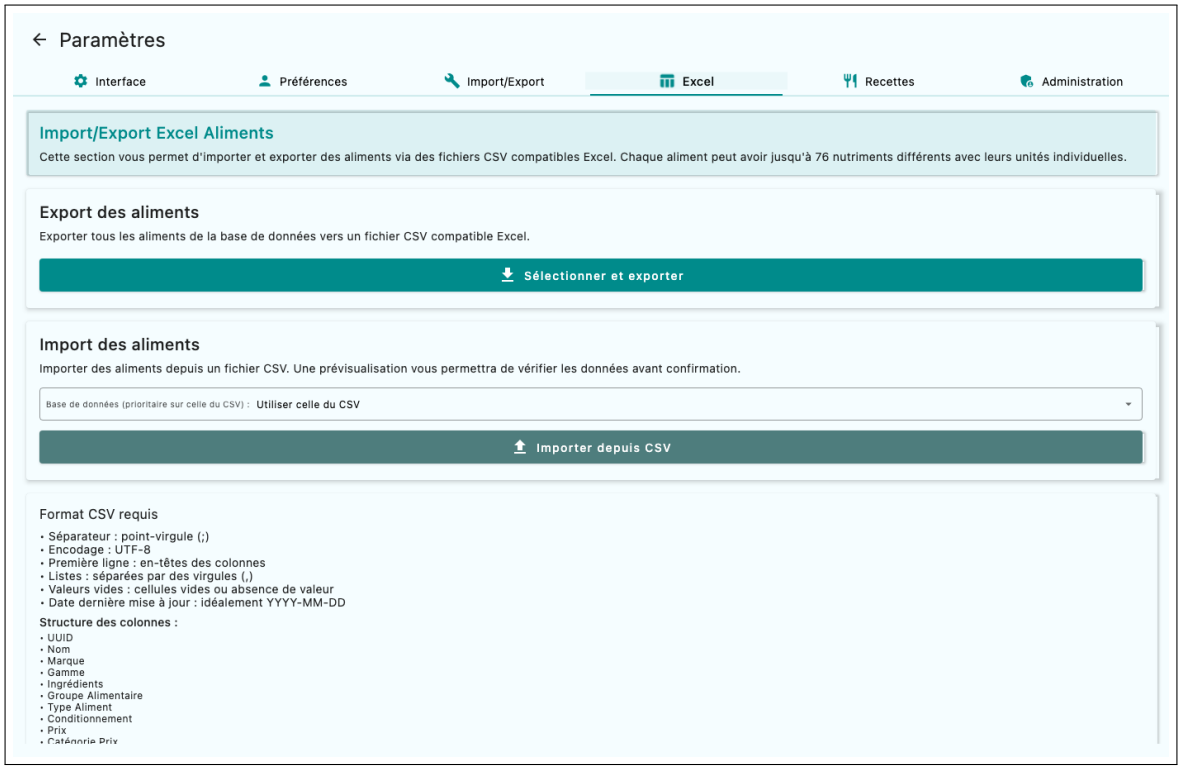



**Figure 31:** Excel export and import of foods.

## 9.4 Recipes

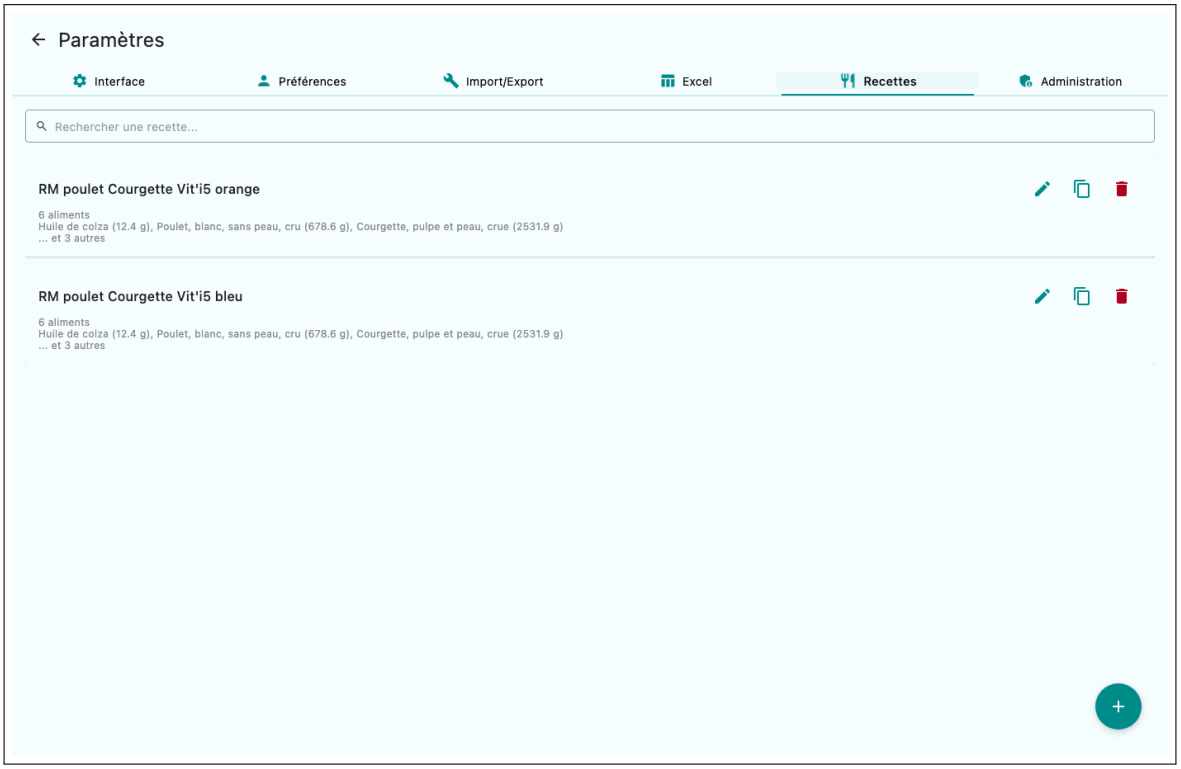



**Figure 32:** Saved recipes.

## 9.5 Administration

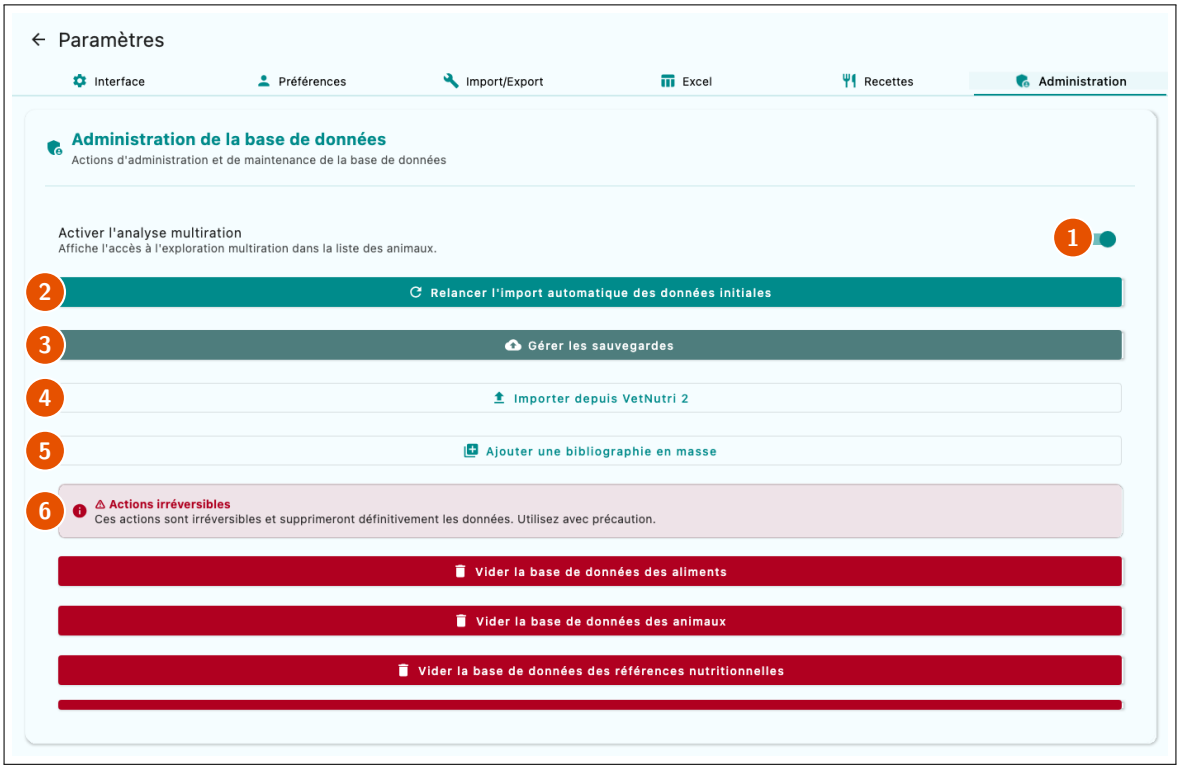



**Figure 33:** Database administration.

1. **Enable multi-ration analysis** shows the **Multi-ration exploration** button in the animal list.
2. Rerun the automatic import of the initial data (foods, references, advice).
3. **Manage backups** (section 9.6).
4. Import the data of a VetNutri 2 installation.
5. Add the same bibliographic reference to many foods.
6. Irreversible actions: clearing a database.

**Caution**

The red **Clear ... database** buttons permanently delete the foods, animals, references, equations or bibliographies. Make a backup before using them.

## 9.6 Backups

**Administration** > **Manage backups** lists the backups with their date, content and size. **New backup** creates one; on each row, one icon restores it and another deletes it.

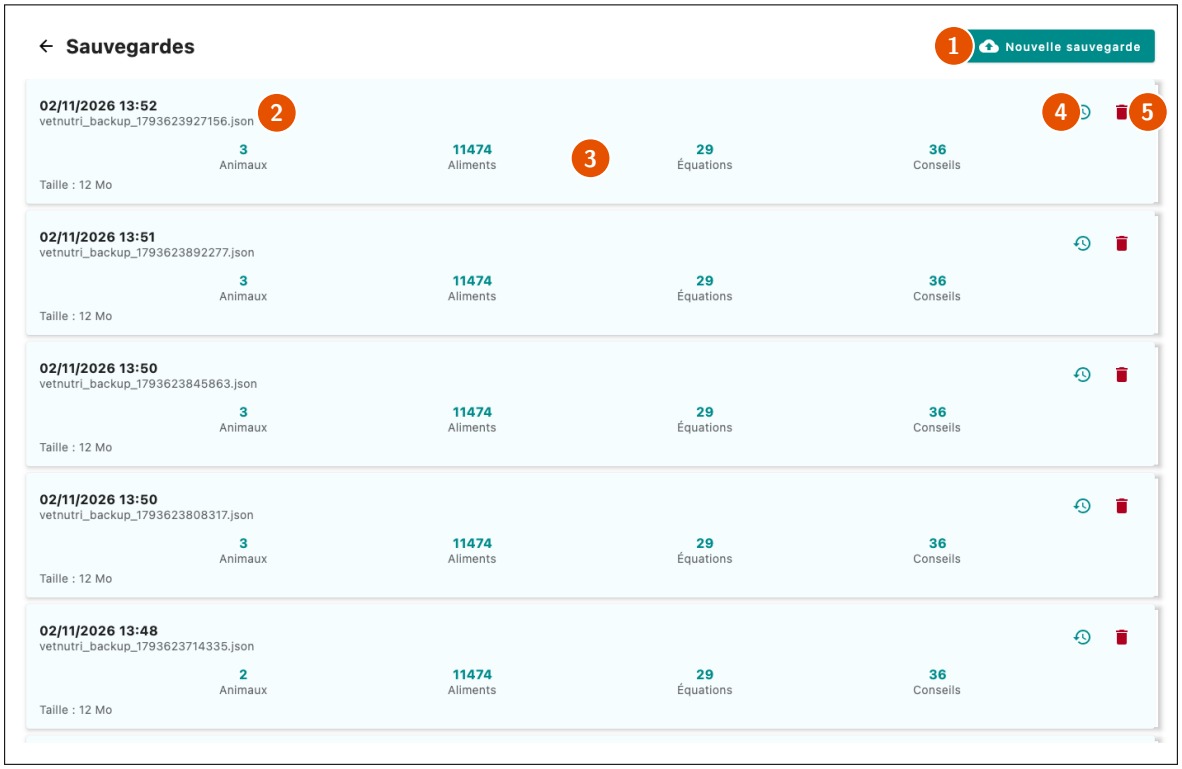



**Figure 34:** Managing backups.

1. **New backup**.
2. The backup's date and file.
3. Its content: animals, foods, equations, advice, and its size.
4. Restore this backup: its data is merged with the current data, after an automatic backup is made first.
5. Delete the backup.

**Key point**

Make a backup before any major operation: bulk import, clearing a database, changing computer. From time to time, copy a backup off the computer.

## 9.7 Cross analysis

**When to use it?** To compare many consultations: tutorials, case series, exams.

Open it with the **Cross analysis** icon, to the left of the settings in the bar of the animal list (figure 5). Filter by species, keywords, exam ID or exercise ID, select the consultations, then open the analysis or the grading.

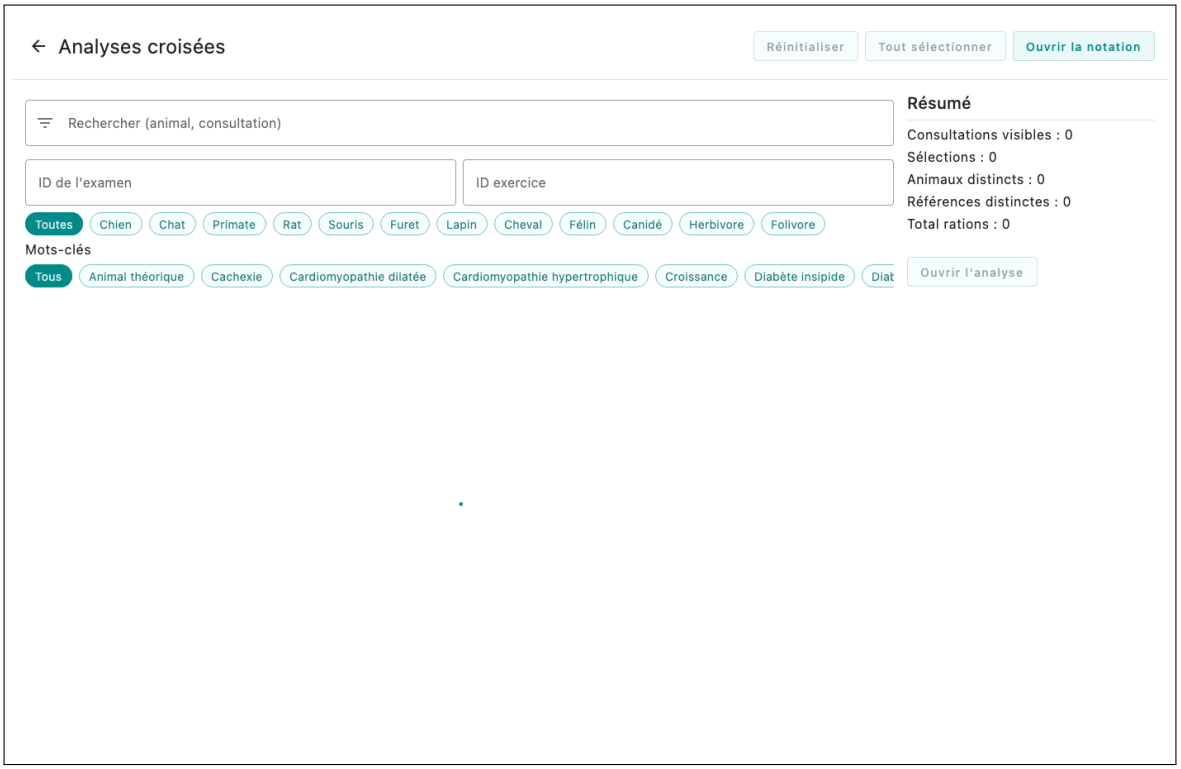



**Figure 35:** Cross analysis.

# A Frequently asked questions

**I cannot save the consultation.**
The **General reference** is missing. Choose a reference in the **Nutritional references** section of the form.

**No reference appears in the list.**
Check the animal's species: only references for its species are offered. Also check that the data has been installed (welcome screen).

**Every nutrient is deficient.**
Look at the energy coverage: if it is far from 100 %, adjust the amount first (section 5).

**The energy requirement looks too high for an overweight animal.**
Calculations use the current weight as long as the ideal weight is not entered. Copy the estimate, or your own value, into **Ideal weight (kg)**.

**The growth curve does not match my puppy.**
Enter its expected adult weight (variable AW) in the consultation, or choose another curve with the button above the graph.

**The multi-ration exploration button does not appear.**
Enable it in **Settings** > **Administration**.

**The energy of a high-fibre food looks low.**
When crude fibre is not entered, total dietary fibre, often much higher, is used in its place in the equation (section 5). Enter the crude fibre in the food's record if you know it.

**A food is missing from the database.**
Create it from the **Food list**, or directly from the food search of a ration (+ at the top right).

# B Glossary

| Term | Definition |
|---|---|
| BCS | Body condition score, from 1 (very thin) to 9 (very obese); 4 and 5 are ideal. |
| BE | The animal's energy requirement: BEE × K, plus energy related to diseases. |
| BEE | Standard energy requirement, calculated by the reference's equation (kcal/day). |
| Coverage | Share of the BE supplied by the ration (%). |
| Crude fibre | Fibre fraction used by the energy equation; total dietary fibre if missing. |
| Current ration | First ration of a consultation, often what the animal already eats. |
| Herd | Record describing a group of animals by groups of individuals. |
| Ideal weight | Weight on which requirements are calculated; the current weight if missing. |
| K, K1 to K5 | Coefficients that adjust the energy requirement (activity, neutering…). |
| Metabolic weight | Weight raised to the power 0.75 ($kg^{0.75}$). |
| NFE | Nitrogen-free extract (digestible carbohydrates), in %; calculated by difference if missing. |
| Observed K | Energy supplied divided by the BEE: the coefficient actually applied. |
| Reference | Set of requirements (minimum, optimum, maximum) for one type of animal. |

# C Quick reference: a consultation in ten moves

1. Animal list: +, or the arrow of an existing animal.
2. **Consultations**: +, or copy the previous consultation.
3. Weight and BCS.
4. Ideal weight: copy the estimate or enter your own value.
5. General reference, plus complementary ones for a disease; confirm.
6. **Rations**: + in **Ration foods**, search, add.
7. Read the energy balance.
8. Quick adjustment, or multi-nutrient adjustment.
9. Review the warnings nutrient by nutrient.
10. **Prescription**: tick the ration, add advice and report, print.